

Geospatial Analysis of Clinico-Genomic Features using Real-World Data from a Cohort of New York City Colorectal Cancer Patients

McKenzie Skrastins

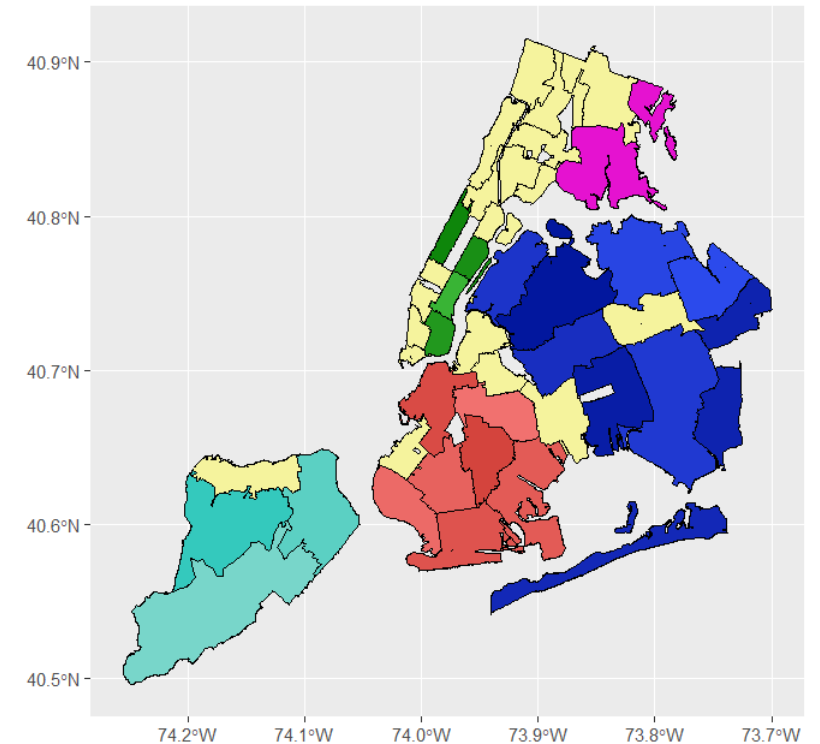
Dr. Sharafudeen Abubakar

Dr. Francisco Sanchez-Vega

August 7, 2025

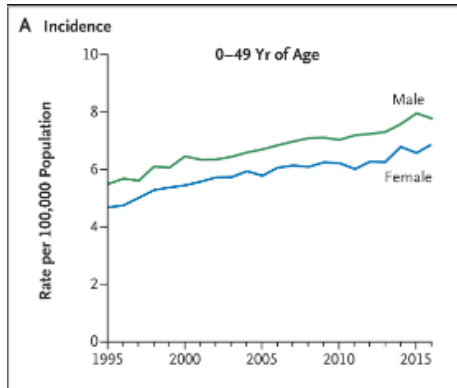


Memorial Sloan Kettering
Cancer Center



Agenda

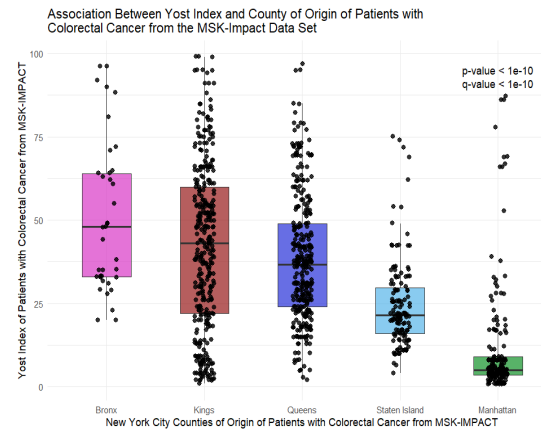
1. Background



2. Methods



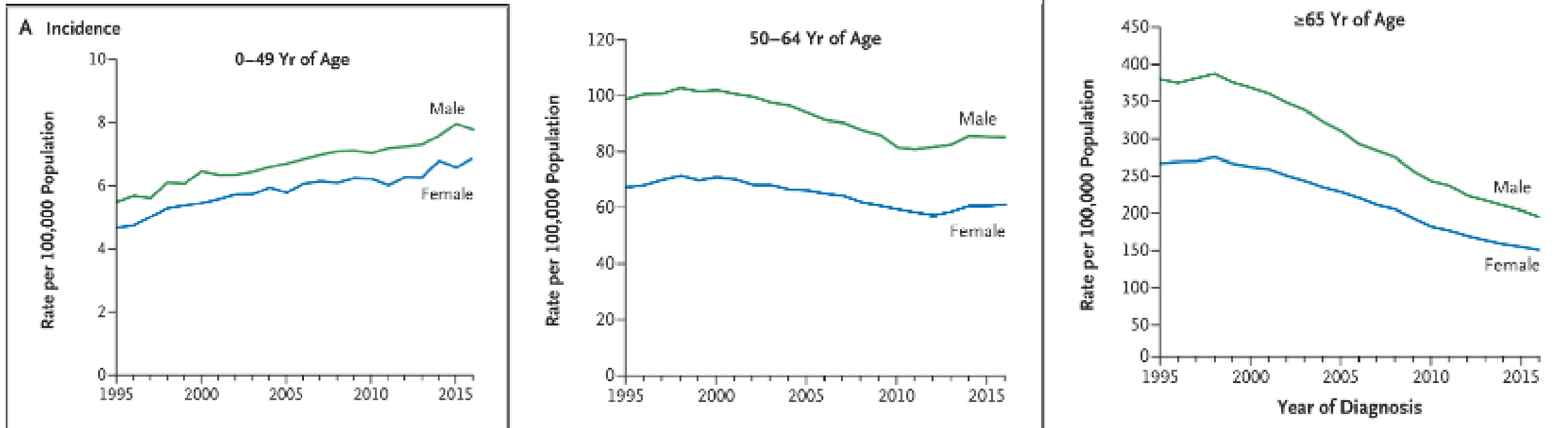
3. Results



4. Conclusion



Background: Colorectal Cancer is a type of cancer that originates in the colon or rectum



Trends in Colorectal Cancer Incidence (1995–2016) in the United States, According to Age and Sex.

Siegel RL, Wagle NS, Cercek A, Smith RA, Jemal A. Colorectal Cancer Statistics, 2023. *CA Cancer J Clin.* 2023; 73(3): 233-254. doi:10.3322/caac.21772

Sinicrope, Frank A. Increasing Incidence of Early-Onset Colorectal Cancer. *New England Journal of Medicine*, Vol 386, No. 16, p 1547-1558, 2022

10.1056/NEJMra2200869

Background: Our investigation of location and colorectal cancer explores variable relationships that most other investigations do not

- Geographic location is a risk for developing cancer, due to exposure



<https://www.amazon.com/develop-com-Android-Location-Pro/dp/B00KQGG2DM>

Background: Our investigation of location and colorectal cancer explores variable relationships that most other investigations do not

- Geographic location is a risk for developing cancer, due to exposure
- Previous investigations:
 - Uses community type as a placeholder for geographic location (Andrilla et al., 2019)



<https://www.amazon.com/ldevelop-com-Android-Location-Pro/dp/B00KQGG2DM>

Andrilla, Holly A. et al. Investigating the Impact of Geographic Location on Colorectal Cancer Stage at Diagnosis: A National Study of the SEER Cancer Registry. The Journal of Rural Health, Vol. 36, Issue 3, p. 316-325, Aug. 2019, <https://doi/10.1111/jrh.12392>

Background: Our investigation of location and colorectal cancer explores variable relationships that most other investigations do not

- Geographic location is a risk for developing cancer, due to exposure
- Previous investigations:
 - Uses community type as a placeholder for geographic location (Andrilla et al., 2019)
 - lack clinical and genomic variables (Buck, 2014)



<https://www.amazon.com/ldevelop-com-Android-Location-Pro/dp/B00KQGG2DM>

Andrilla, Holly A. et al. Investigating the Impact of Geographic Location on Colorectal Cancer Stage at Diagnosis: A National Study of the SEER Cancer Registry. The Journal of Rural Health, Vol. 36, Issue 3, p. 316-325, Aug. 2019, <https://doi.org/10.1111/jrh.12392>

Kyle D. Buck. Modelling of geographic cancer risk factor disparities in US counties. Applied Geography, Volume 75, 2016, Pages 28-35, ISSN 0143-6228, <https://doi.org/10.1016/j.apgeog.2016.08.001>.

Methods: The primary datasets used were MSK-IMPACT and CRC Disparities

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MSK data set of
sequenced tumors

95,000 patients

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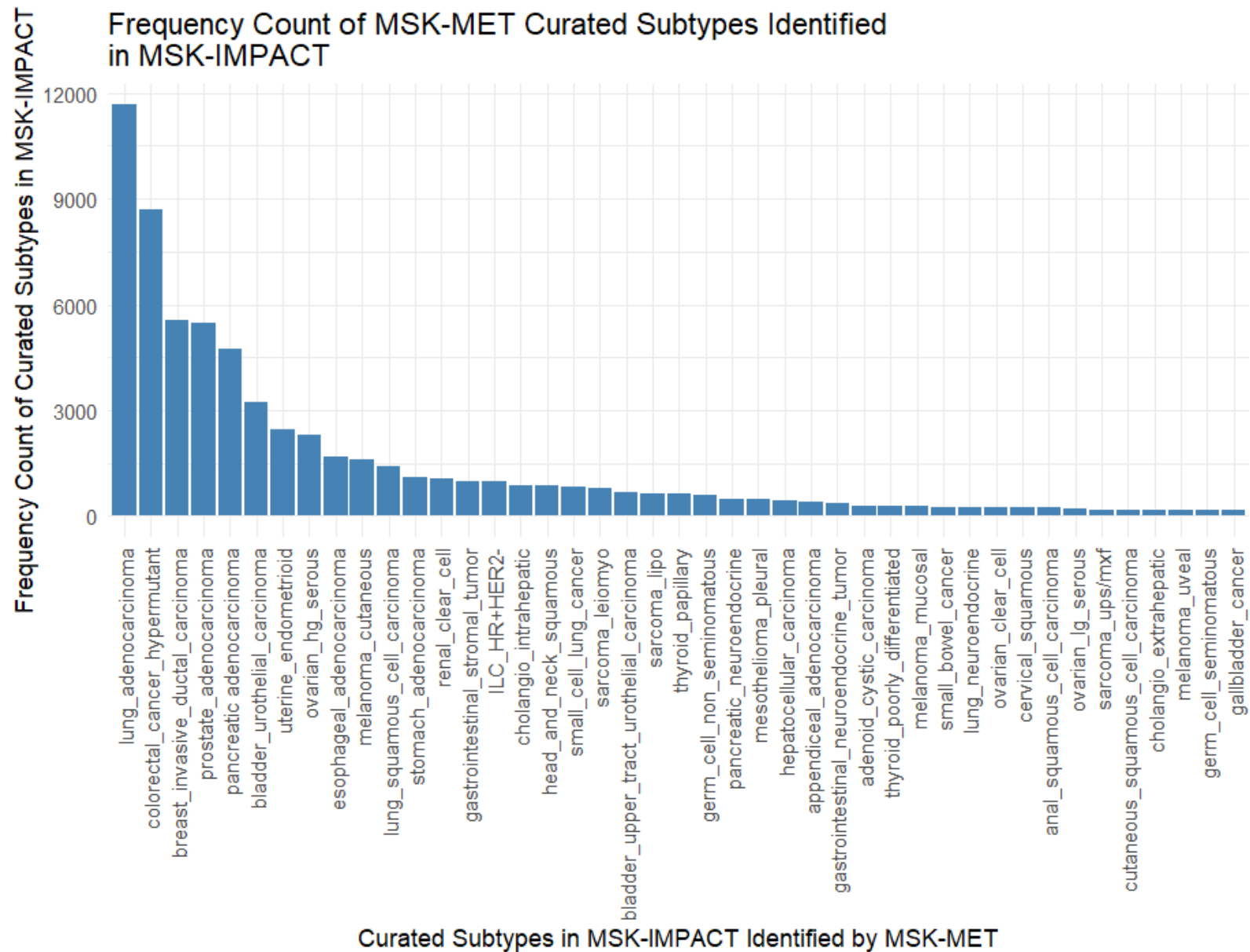
Nguyen B., et al.
Cell vol. 185, 3
2022

MSK-MET

Curated Subtypes
label

65,000 patients

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MSK-IMPACT



MSK data set of
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Figure 1. Overview of the MSK-MET cohort.
Nguyen B., et al.
Cell vol. 185, 3
2022

MSK-MET

Curated Subtypes
label

65,000 patients

Geographic Data for CRC



5,500 patients

Methods: The primary datasets used were MSK-IMPACT and CRC Disparities

Specified Area



1,200 patients

Filter down to
NYC

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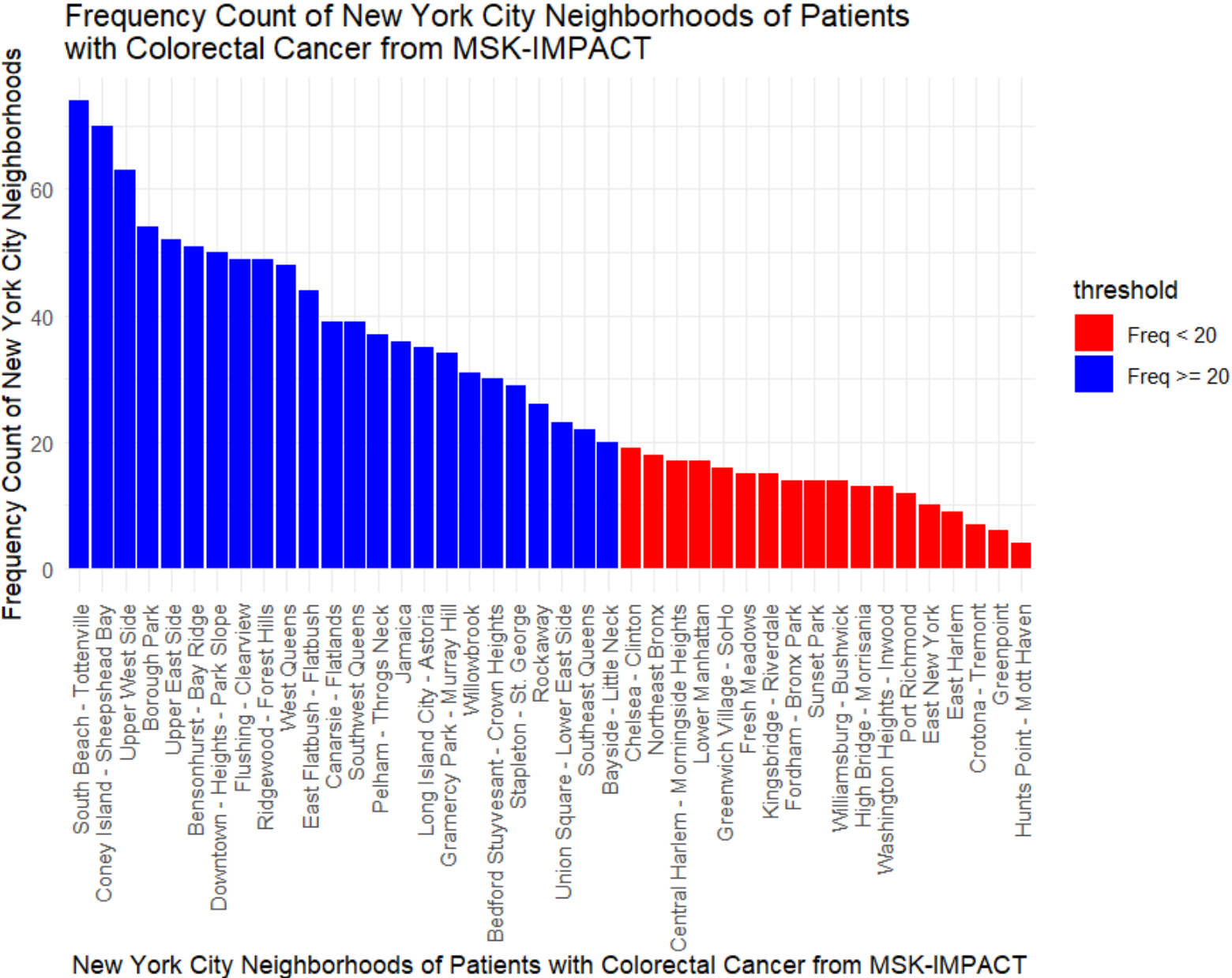
1,200 patients

Map zip codes to
neighborhood and
county level

Methods: Zip codes were mapped to using the United Hospital Fund Codes

UHF code	Neighborhood name	Zip codes
Bronx		
101	Kingsbridge - Riverdale	10463, 10471
102	Northeast Bronx	10466, 10469, 10470, 10475
103	Fordham - Bronx Park	10458, 10467, 10468
104	Pelham - Throgs Neck	10461, 10462, 10464, 10465, 10472, 10473
105	Crotona - Tremont	10453, 10457, 10460
106	High Bridge - Morrisania	10451, 10452, 10456
107	Hunts Point - Mott Haven	10454, 10455, 10459, 10474
Brooklyn		
201	Greenpoint	11211, 11222
202	Downtown - Heights - Park Slope	11201, 11205, 11215, 11217, 11231
203	Bedford Stuyvesant - Crown Heights	11213, 11212, 11216, 11233, 11238
204	East New York	11207, 11208
205	Sunset Park	11220, 11232
206	Borough Park	11204, 11218, 11219, 11230
207	East Flatbush - Flatbush	11203, 11210, 11225, 11226
208	Canarsie - Flatlands	11234, 11236, 11239
209	Bensonhurst - Bay Ridge	11209, 11214, 11228
210	Coney Island - Sheepshead Bay	11223, 11224, 11229, 11235
211	Williamsburg - Bushwick	11206, 11221, 11237

Methods: Any neighborhood that did not have at least 20 patients was removed from the neighborhood level analysis



Methods: The primary datasets used were MSK-IMPACT and CRC Disparities

Specified Area



1,200 patients

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Map zip codes to
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**CRC
Disparities**

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Level of Analysis



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Map zip codes to
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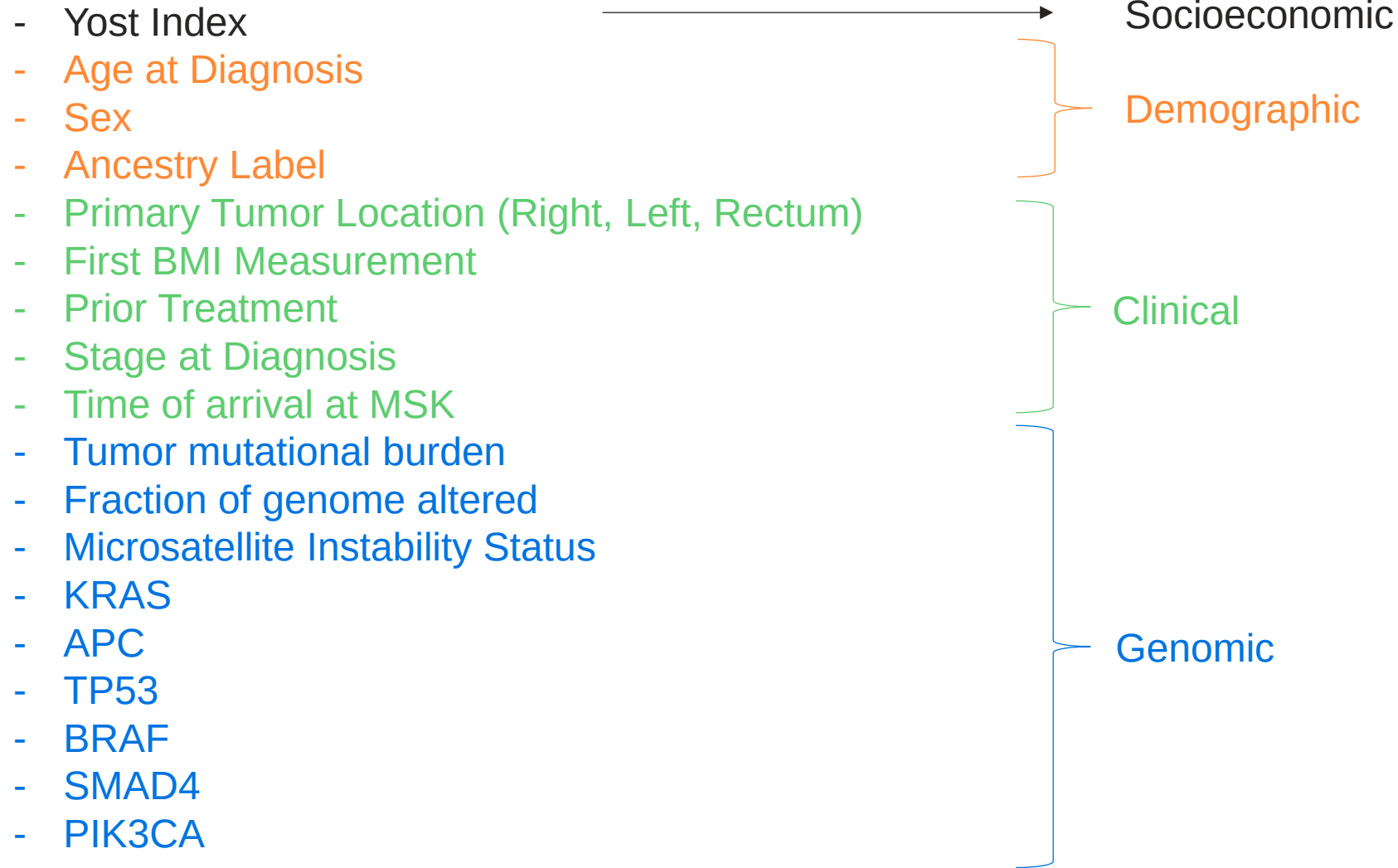
CRC Disparities



1,200 patients

Append relevant
variables

Methods: A variety of variables were used in the analysis



Methods: Ancestry Labels and Yost Index

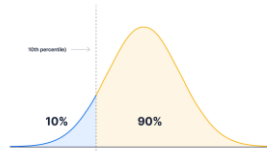
Ancestry Labels



Arora, Kanika et al. 2022

>80%, otherwise
ADM

Yost Index



0 = most affluent
100 = least affluent

Inferred from zip
code data

Arora K, Tran TN, Kemel Y, Mehine M, Liu YL, Nandakumar S, Smith SA, Brannon AR, Ostrovnaya I, Stopsack KH, Razavi P, Safonov A, Rizvi HA, Hellmann MD, Vijai J, Reynolds TC, Fagin JA, Carrot-Zhang J, Offit K, Solit DB, Ladanyi M, Schultz N, Zehir A, Brown CL, Stadler ZK, Chakravarty D, Bandlamudi C, Berger MF. Genetic Ancestry Correlates with Somatic Differences in a Real-World Clinical Cancer Sequencing Cohort. *Cancer Discov.* 2022 Nov 2;12(11):2552-2565. doi: 10.1158/2159-8290.CD-22-0312. PMID: 36048199; PMCID: PMC9633436.

Methods: Univariate analysis was conducted using the Kruskal-Wallis and χ^2

Neighborhood/County vs Variable

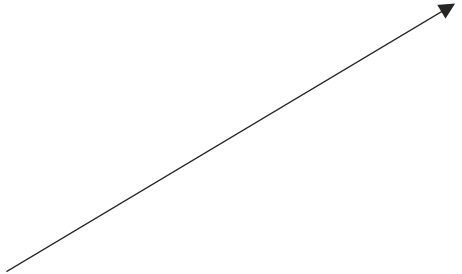
Methods: Univariate analysis was conducted using the Kruskal-Wallis and χ^2

Kruskal-Wallis for continuous vs categorical across all neighborhoods/counties

H0: The distribution of our variable is the same across all neighborhoods/counties

Ha: At least one county/neighborhood has a distribution of the variable that differs from the others

Neighborhood/County vs Variable



Methods: Univariate analysis was conducted using the Kruskal-Wallis and χ^2

Neighborhood/County vs Variable

```
graph LR; A[Neighborhood/County vs Variable] --> B[Kruskal-Wallis for continuous vs categorical across all neighborhoods/counties]; A --> C[Chi^2 for categorical vs categorical across all neighborhoods/counties];
```

Kruskal-Wallis for continuous vs categorical across all neighborhoods/counties

H0: The median of the continuous variable is the same across all neighborhoods/counties

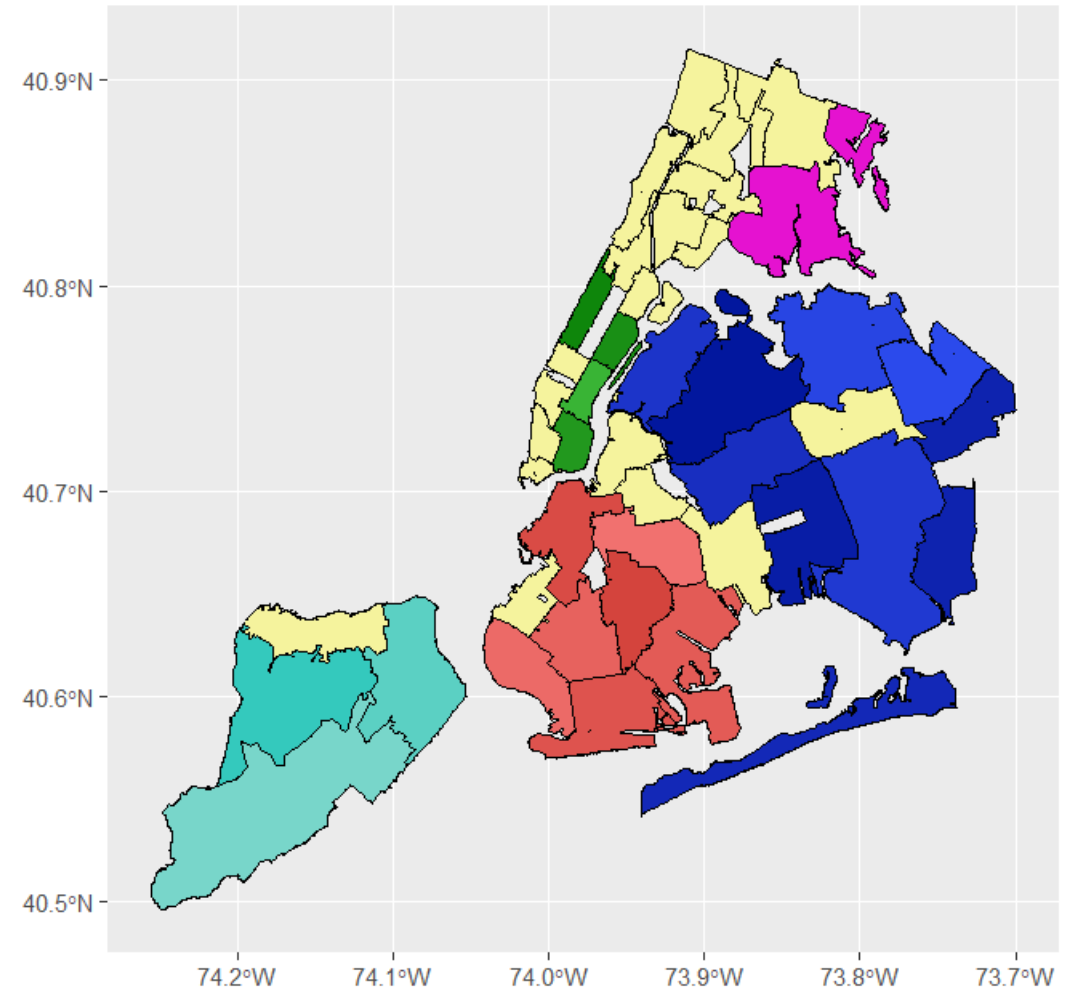
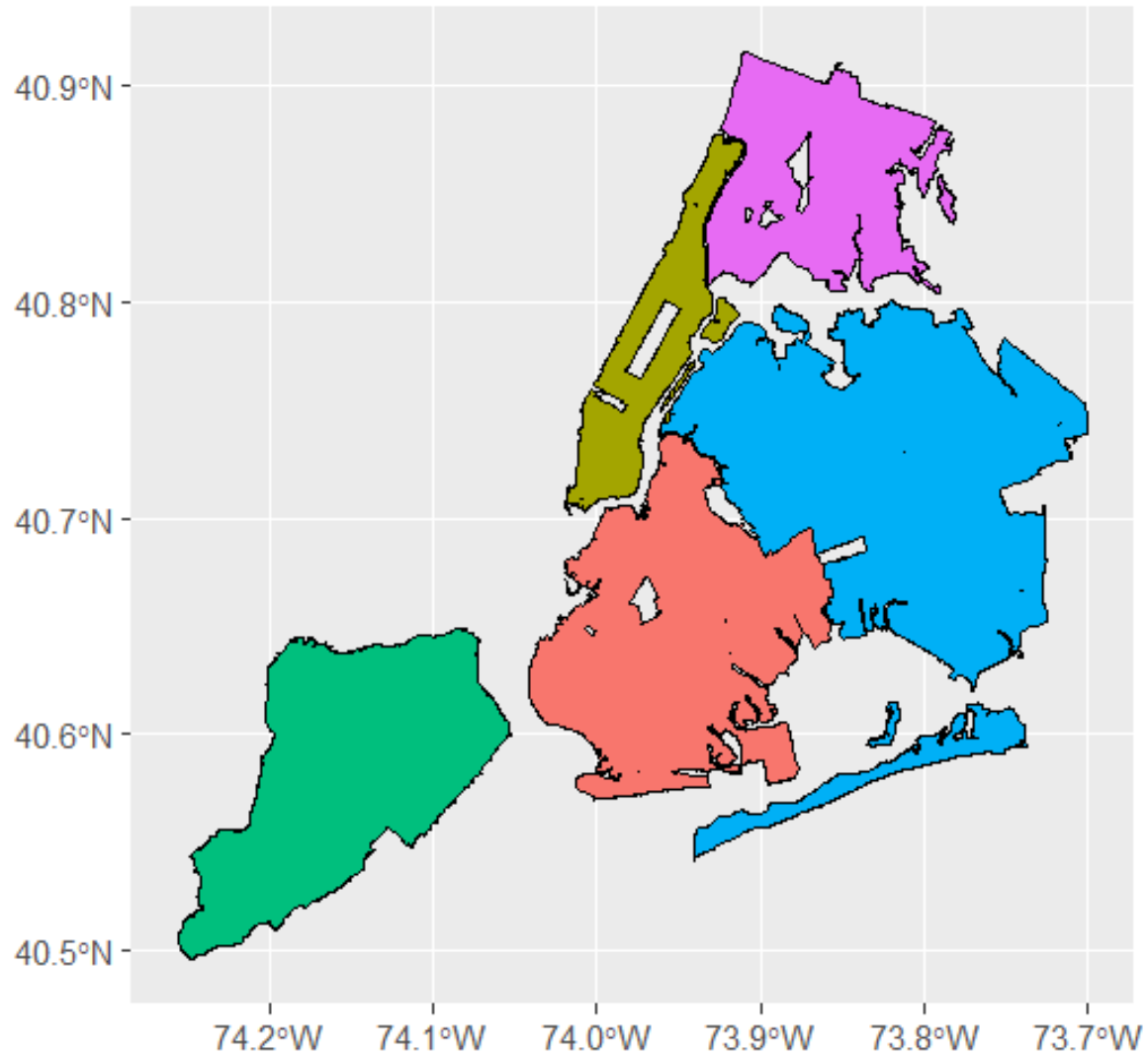
H_a: The median is not the same across all neighborhoods/counties

χ^2 for categorical vs categorical across all neighborhoods/counties

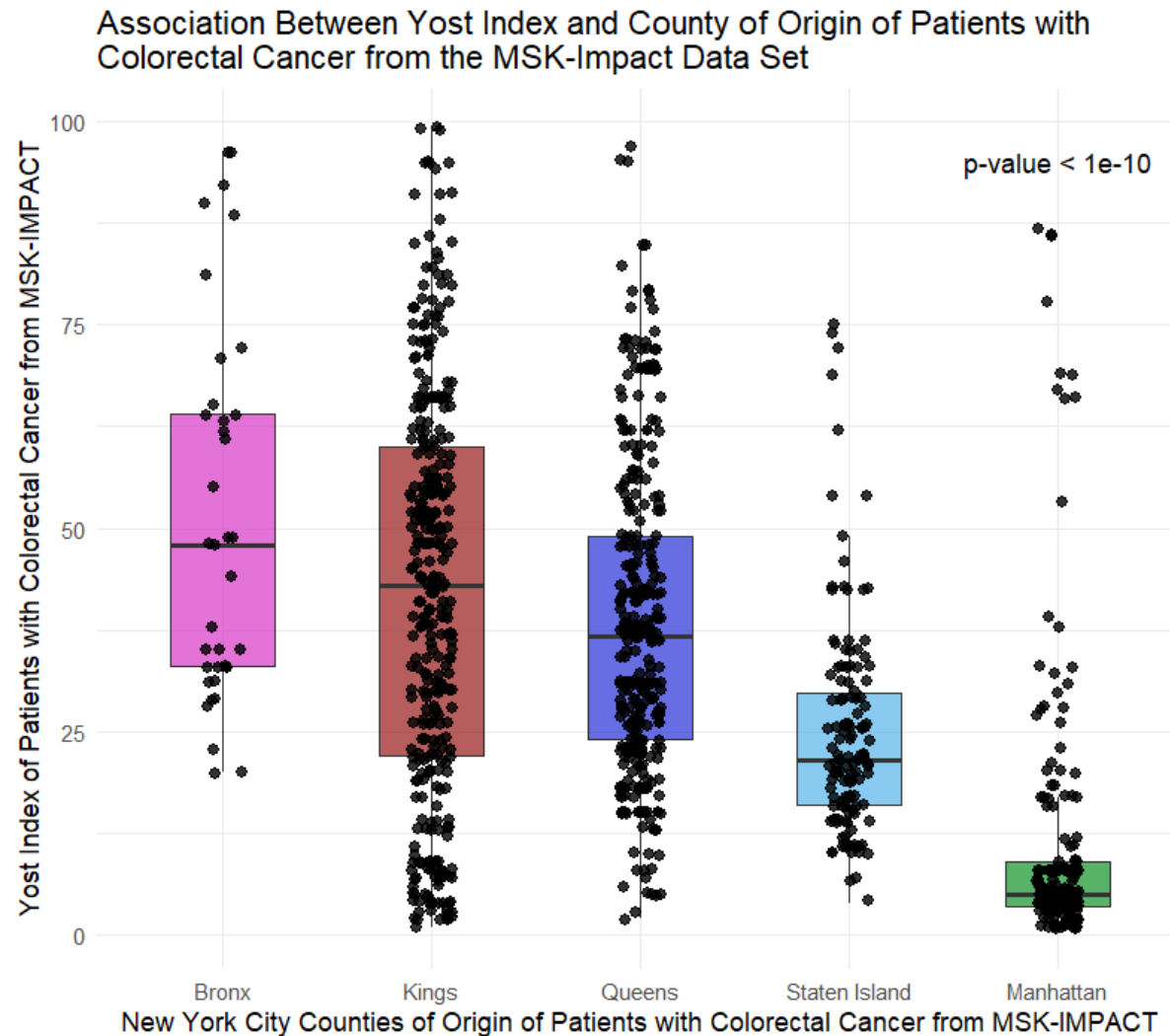
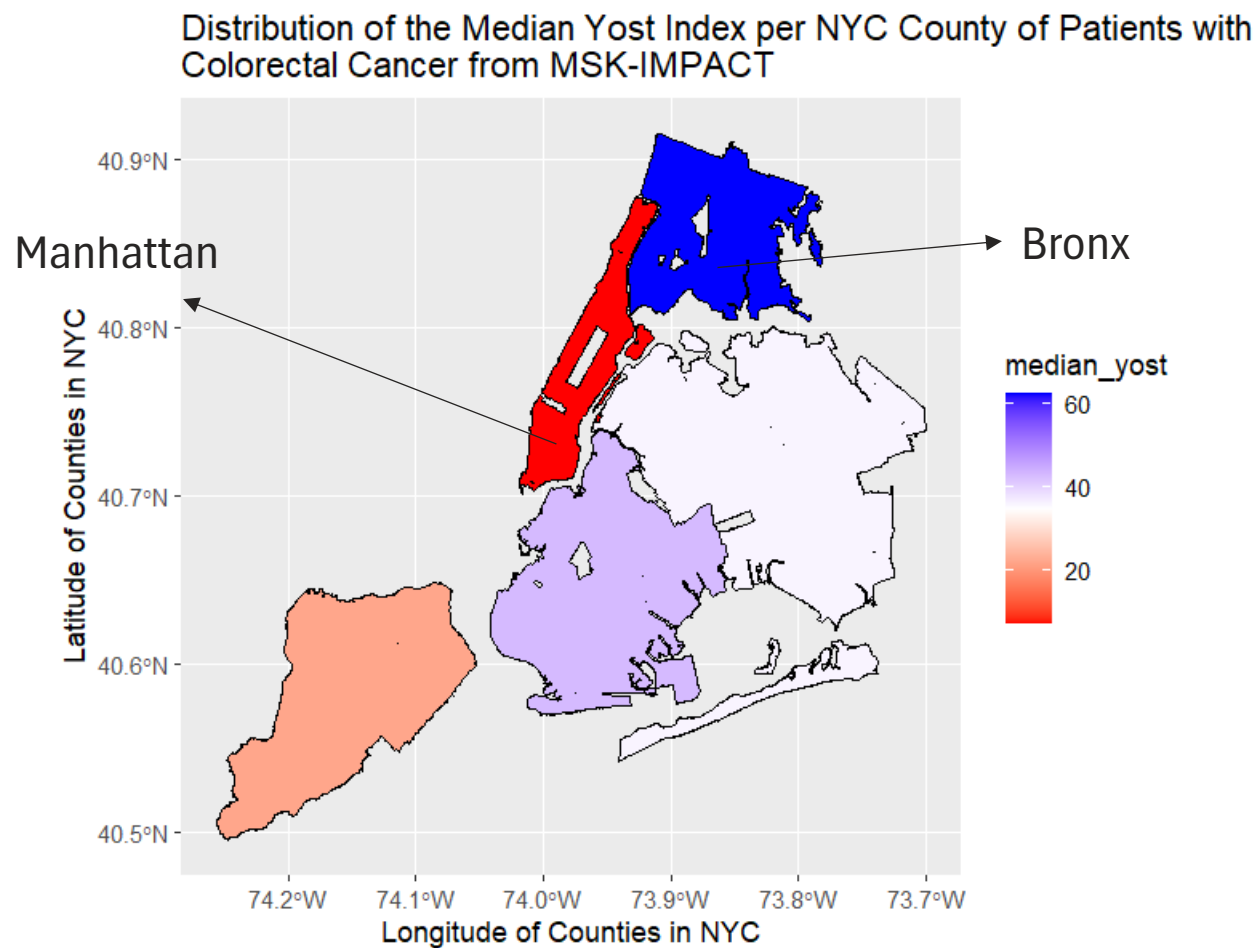
H0: The variable is independent of county/neighborhood

H_a: The variable is statistically dependent on county/neighborhood

Results: Baseline maps of counties and neighborhoods in NYC, the latter being representing neighborhoods with a freq. count >20

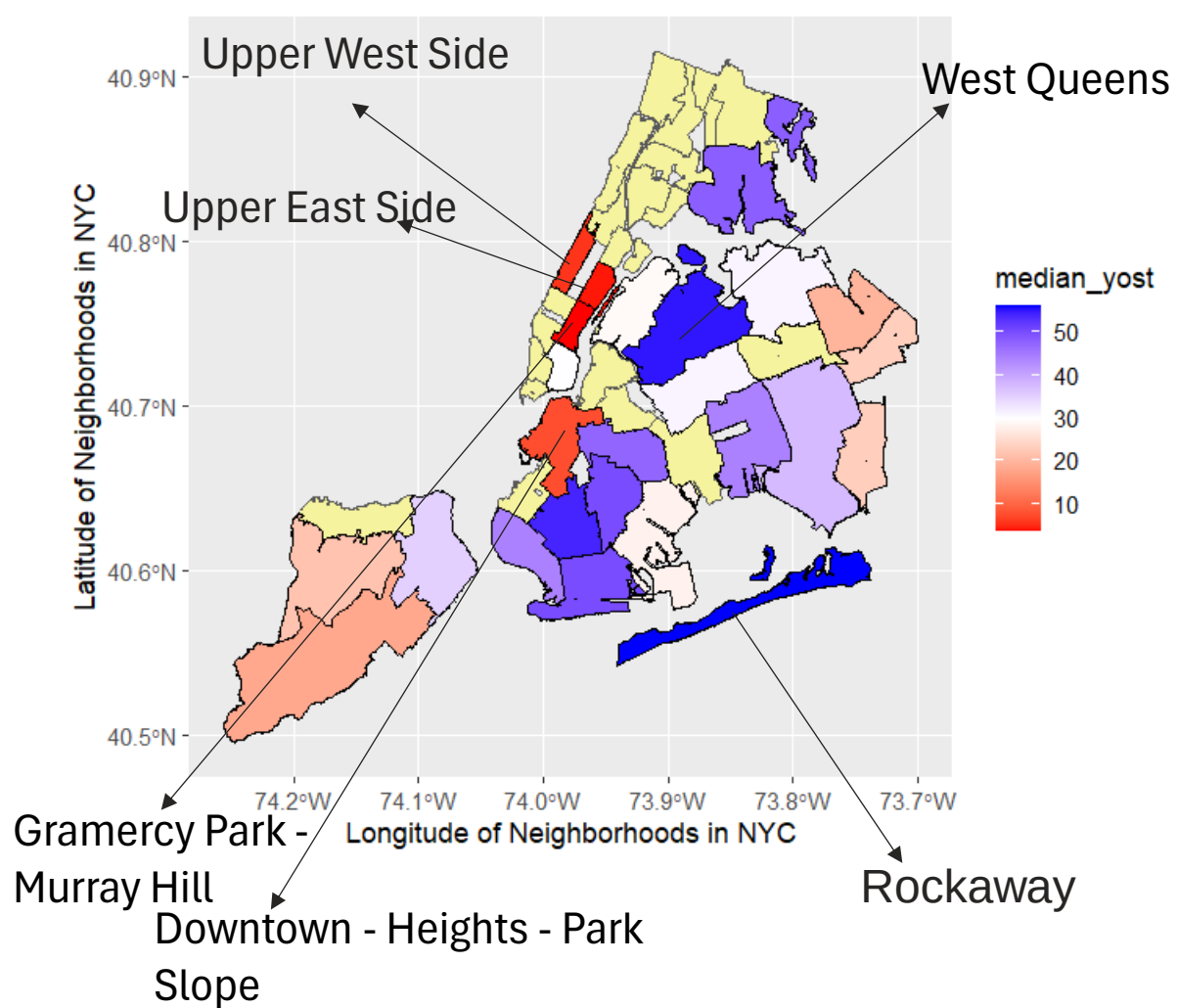


Results: The differences in Yost Index were statistically significant across counties

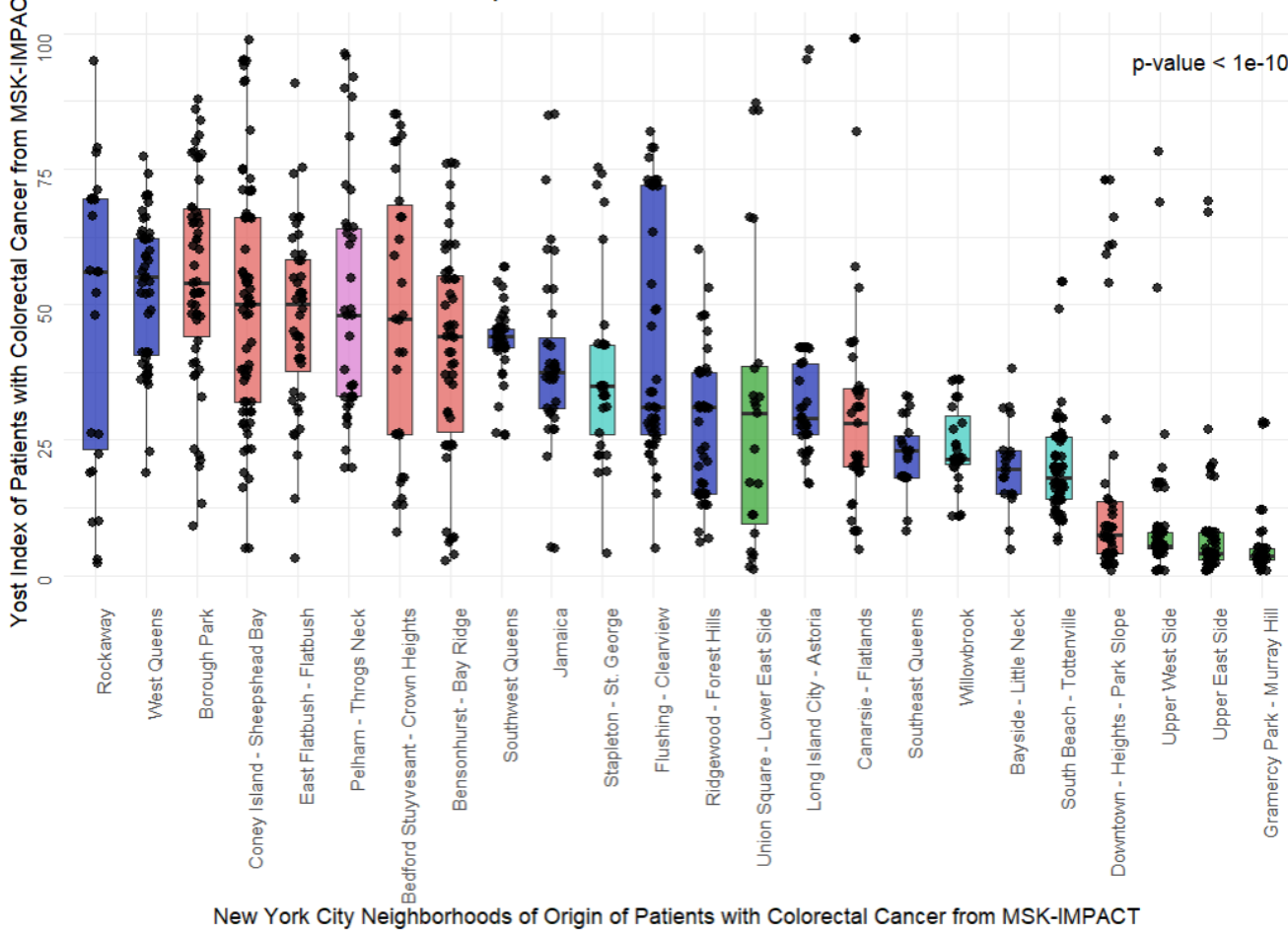


Results: The differences in Yost Index were statistically significant across neighborhoods

Distribution of the Median Yost Index per NYC Neighborhood of Patients with Colorectal Cancer from MSK-IMPACT

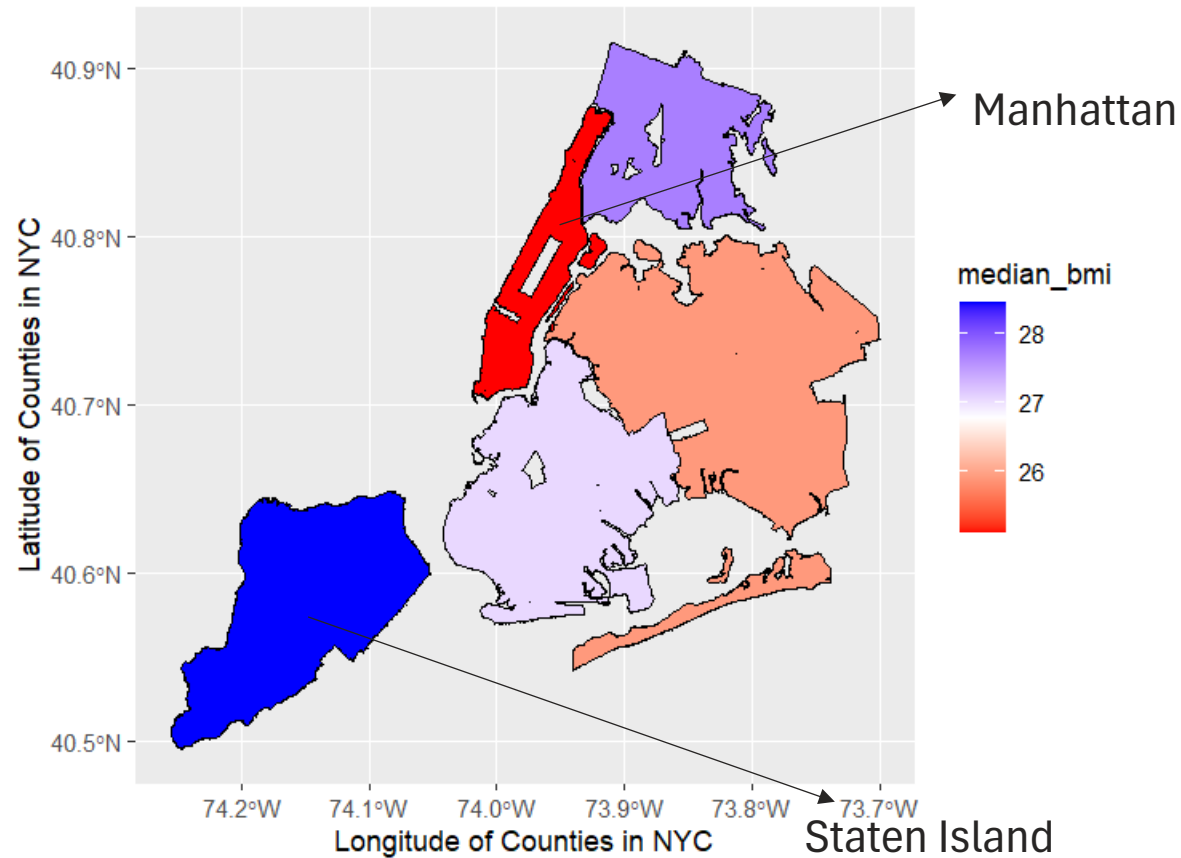


Association Between Yost Index and Neighborhood of Origin of Patients with Colorectal Cancer from the MSK-Impact Data Set

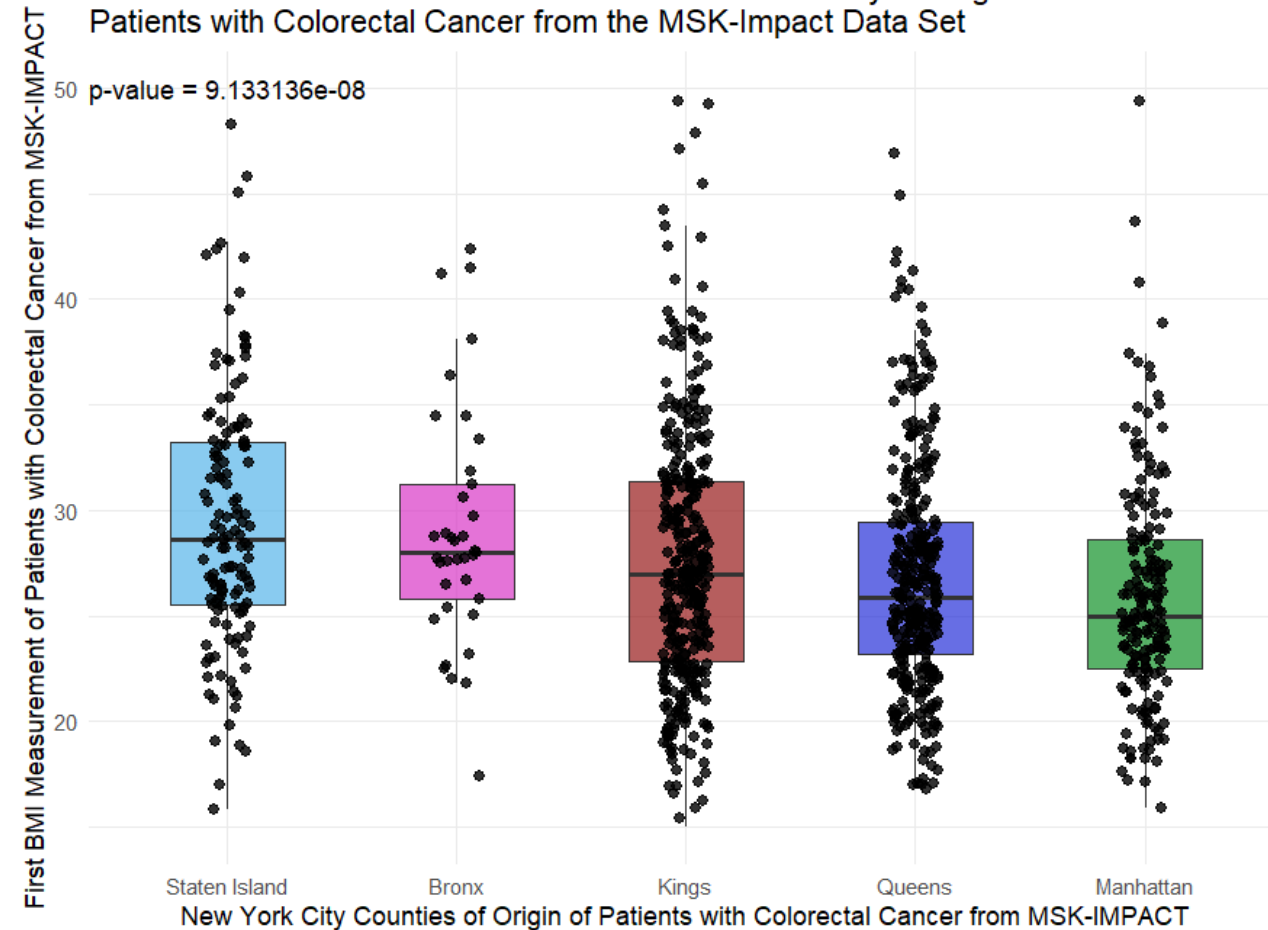


Results: The differences in First BMI Measurement were statistically significant across counties

Distribution of the Median First BMI Measurement per NYC County of Patients with Colorectal Cancer from MSK-IMPACT

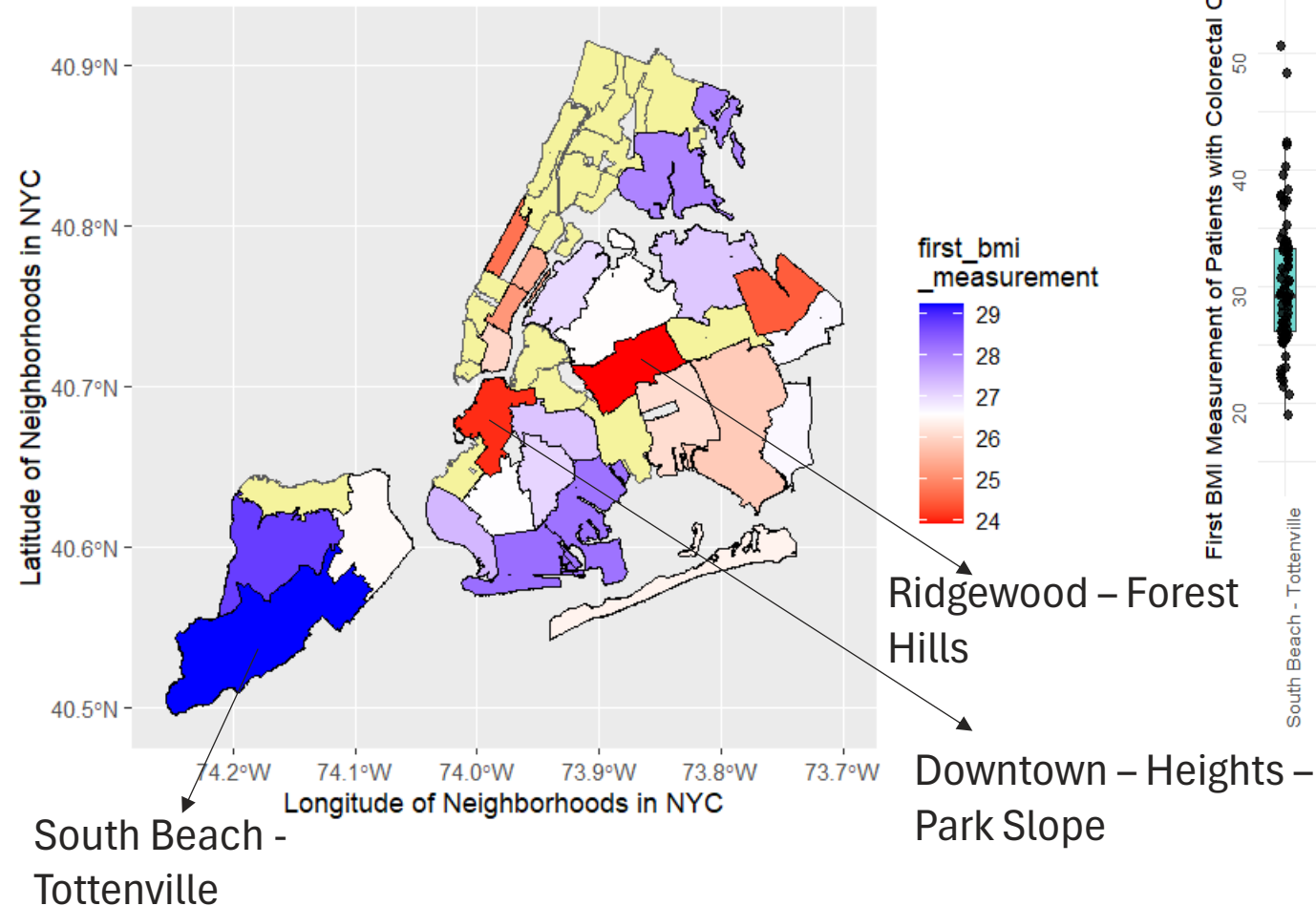


Association Between First BMI Measurement and County of Origin of Patients with Colorectal Cancer from the MSK-Impact Data Set

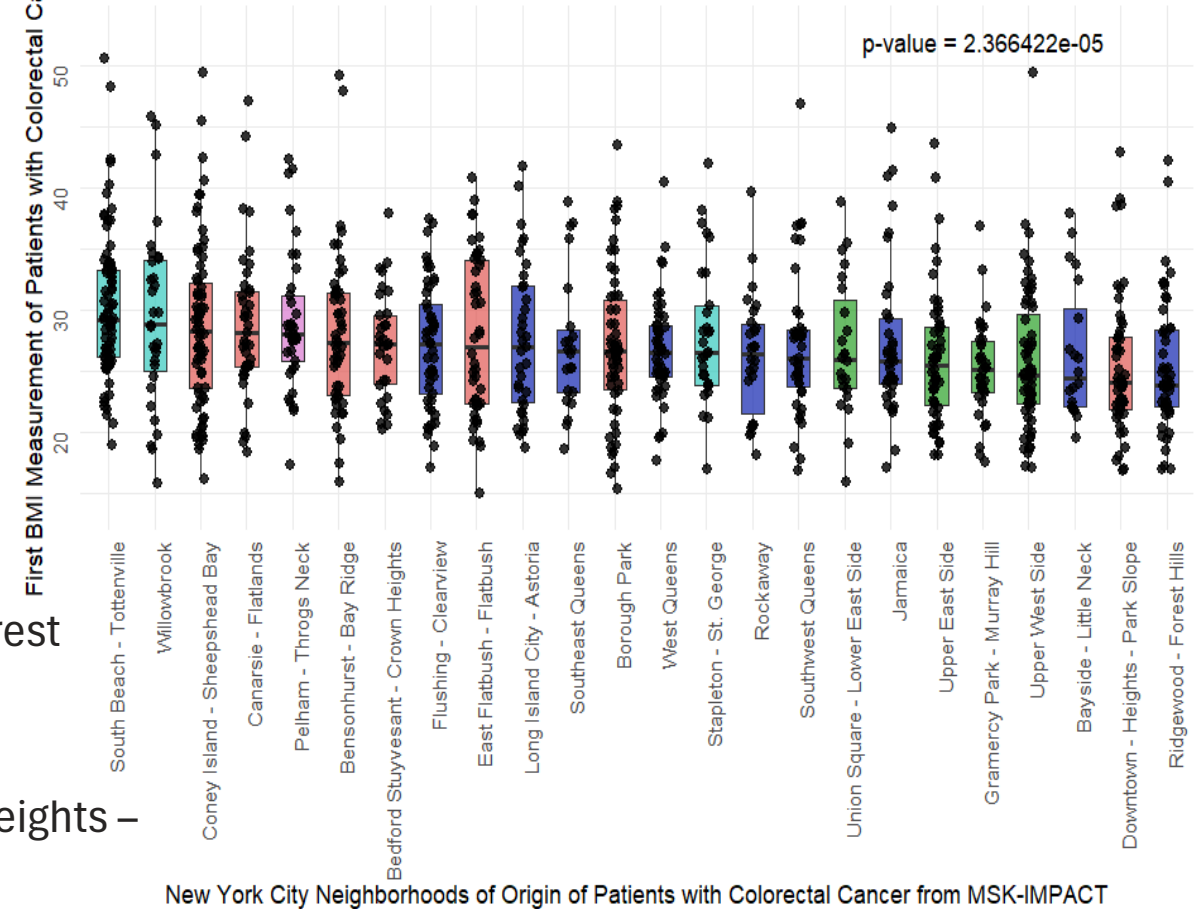


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Distribution of the Median First BMI Measurement per NYC Neighborhood of Patients with Colorectal Cancer from MSK-IMPACT

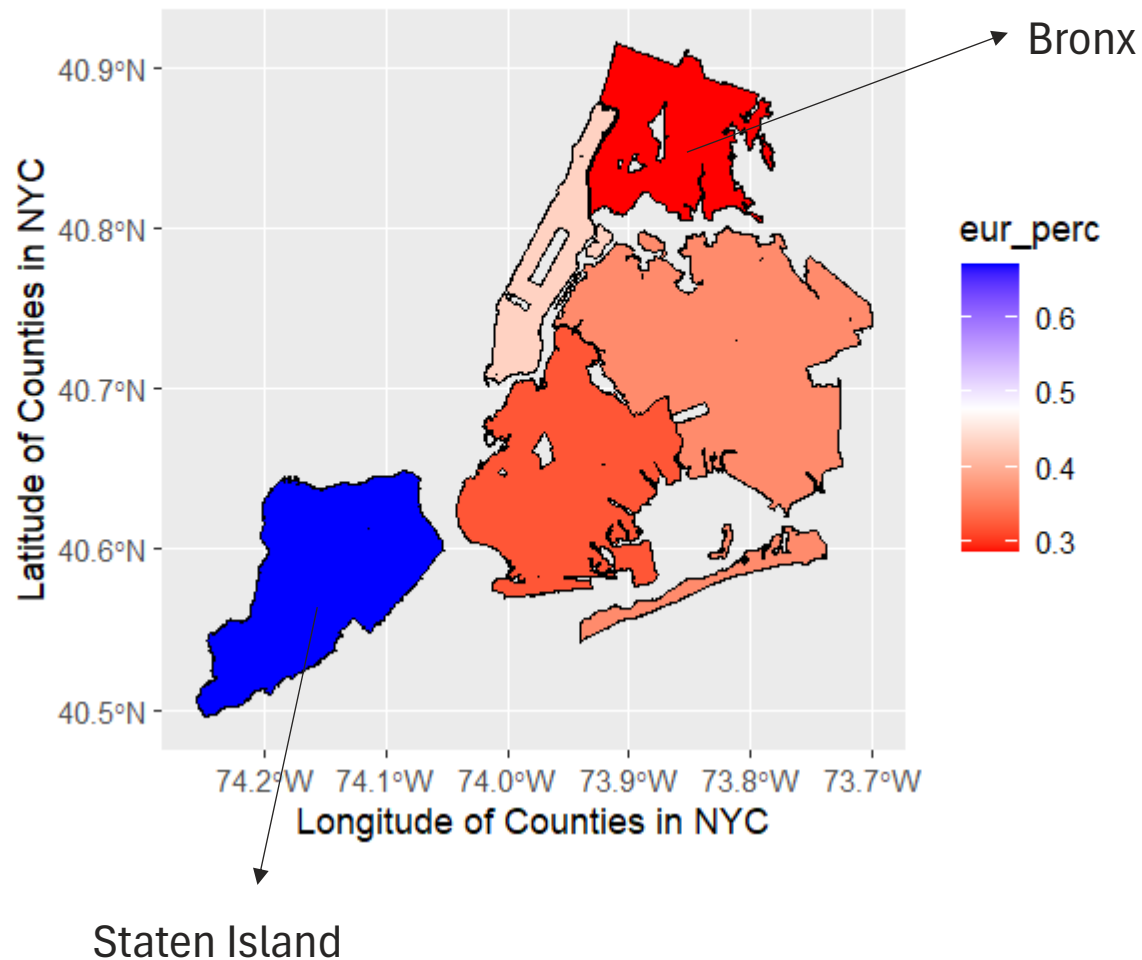


Association Between First BMI Measurement and Neighborhood of Origin of Patients with Colorectal Cancer from the MSK-Impact Data Set

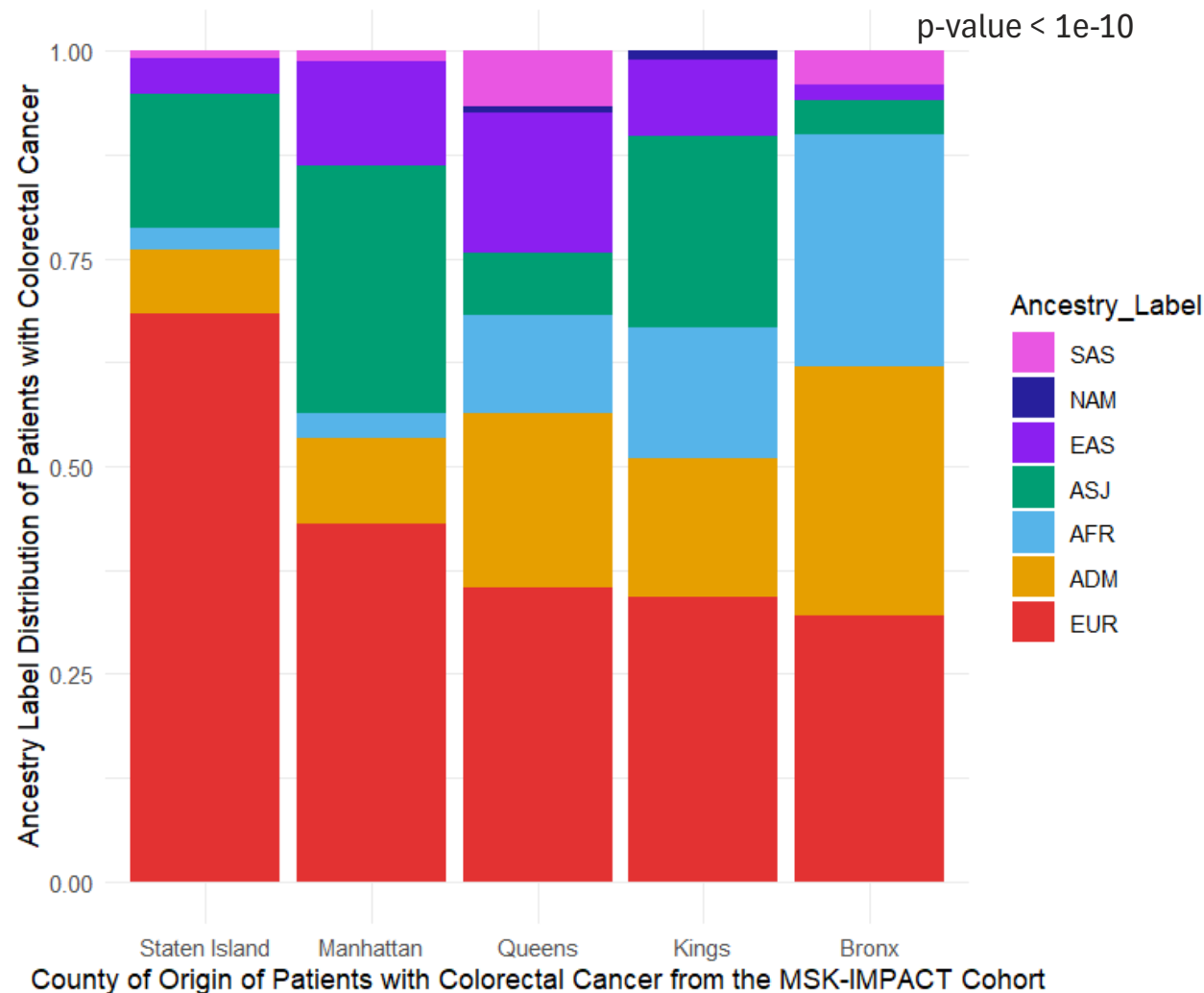


Results: The variation in Ancestry Labels were statistically significant across counties

Distribution of European Patients with Colorectal Cancer from MSK-IMPACT per NYC County

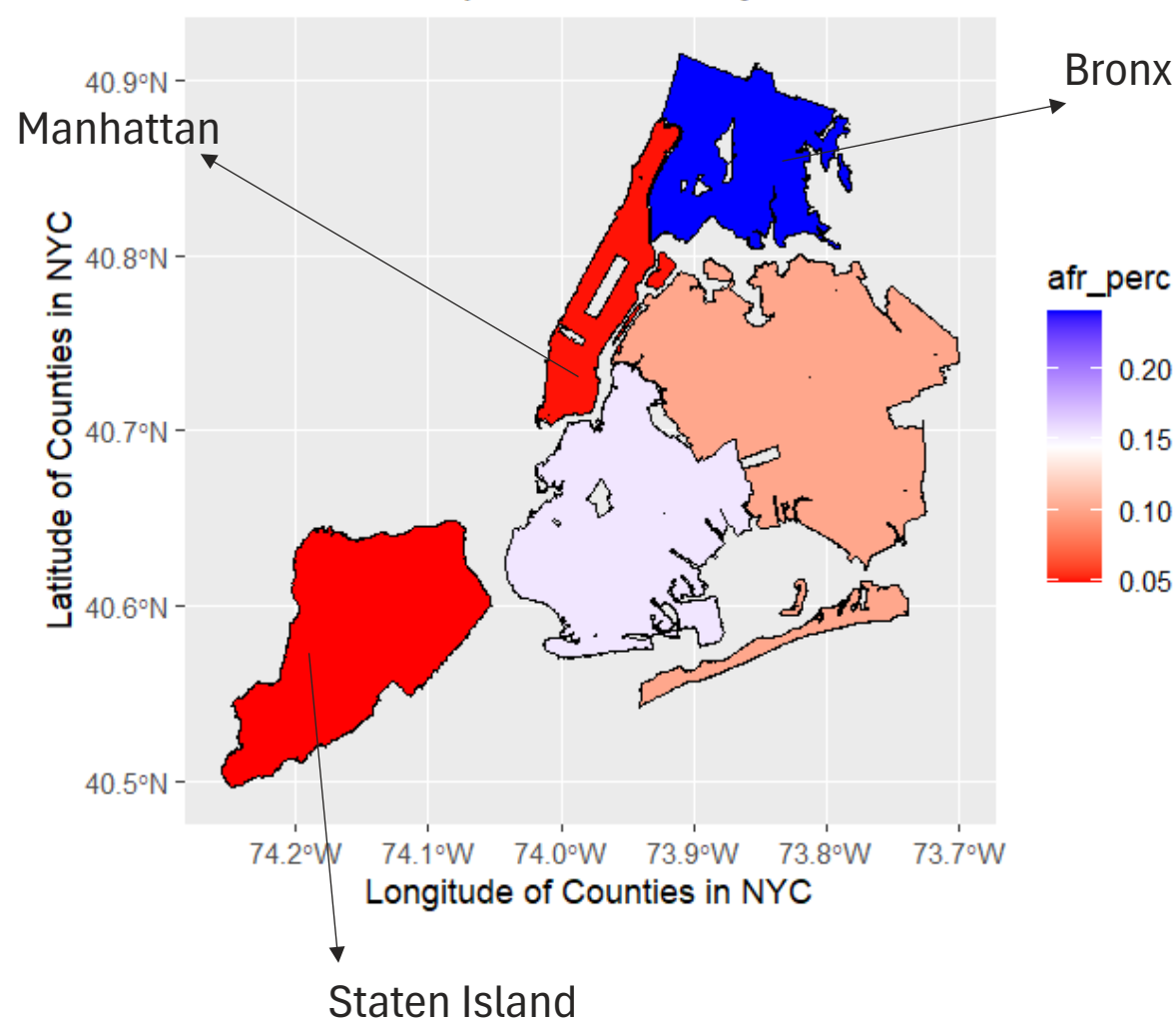


Ancestry Label Distribution of Patients with Colorectal Cancer From MSK-IMPACT by New York City County

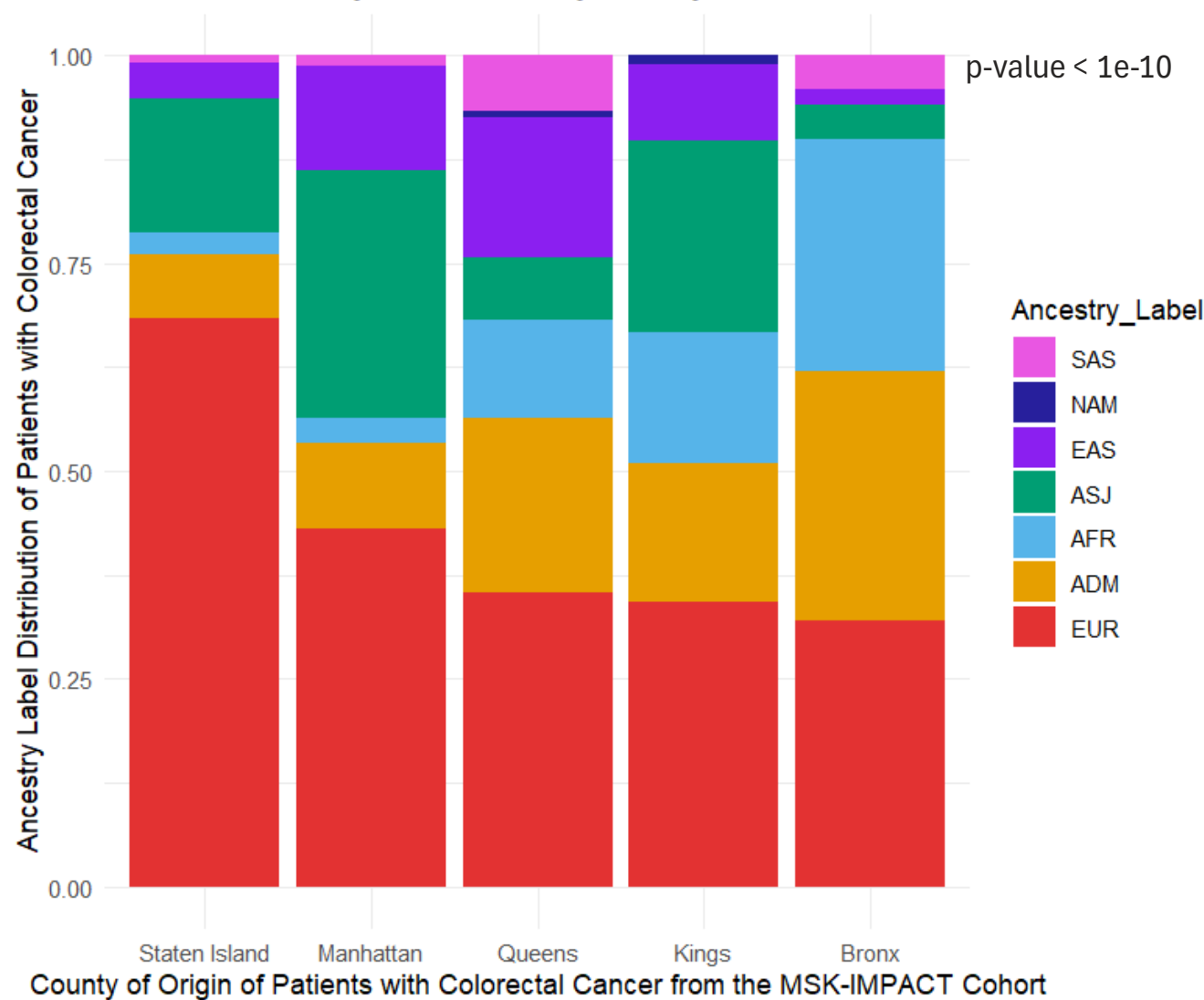


Results: The variation in Ancestry Labels were statistically significant across counties

Distribution of African Patients with Colorectal Cancer from MSK-IMPACT per NYC County

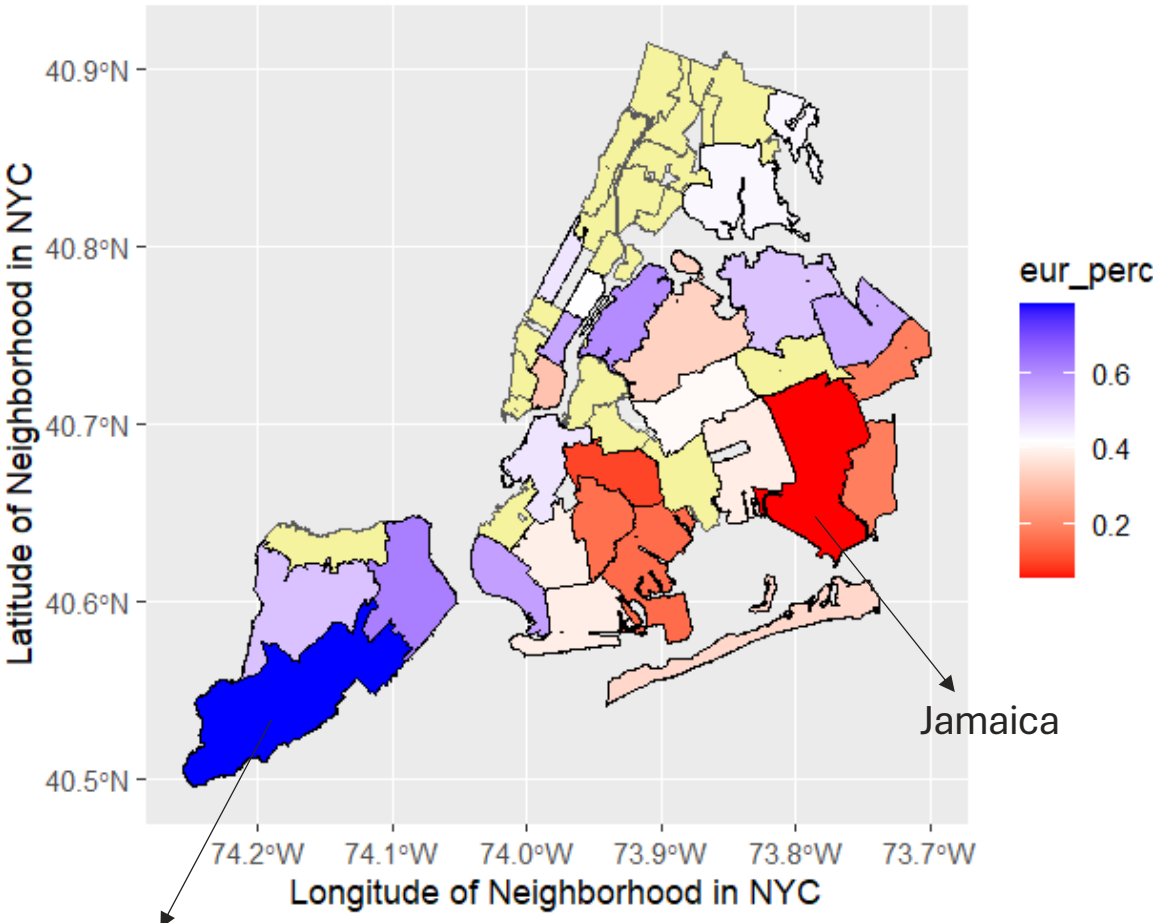


Ancestry Label Distribution of Patients with Colorectal Cancer From MSK-IMPACT by New York City County



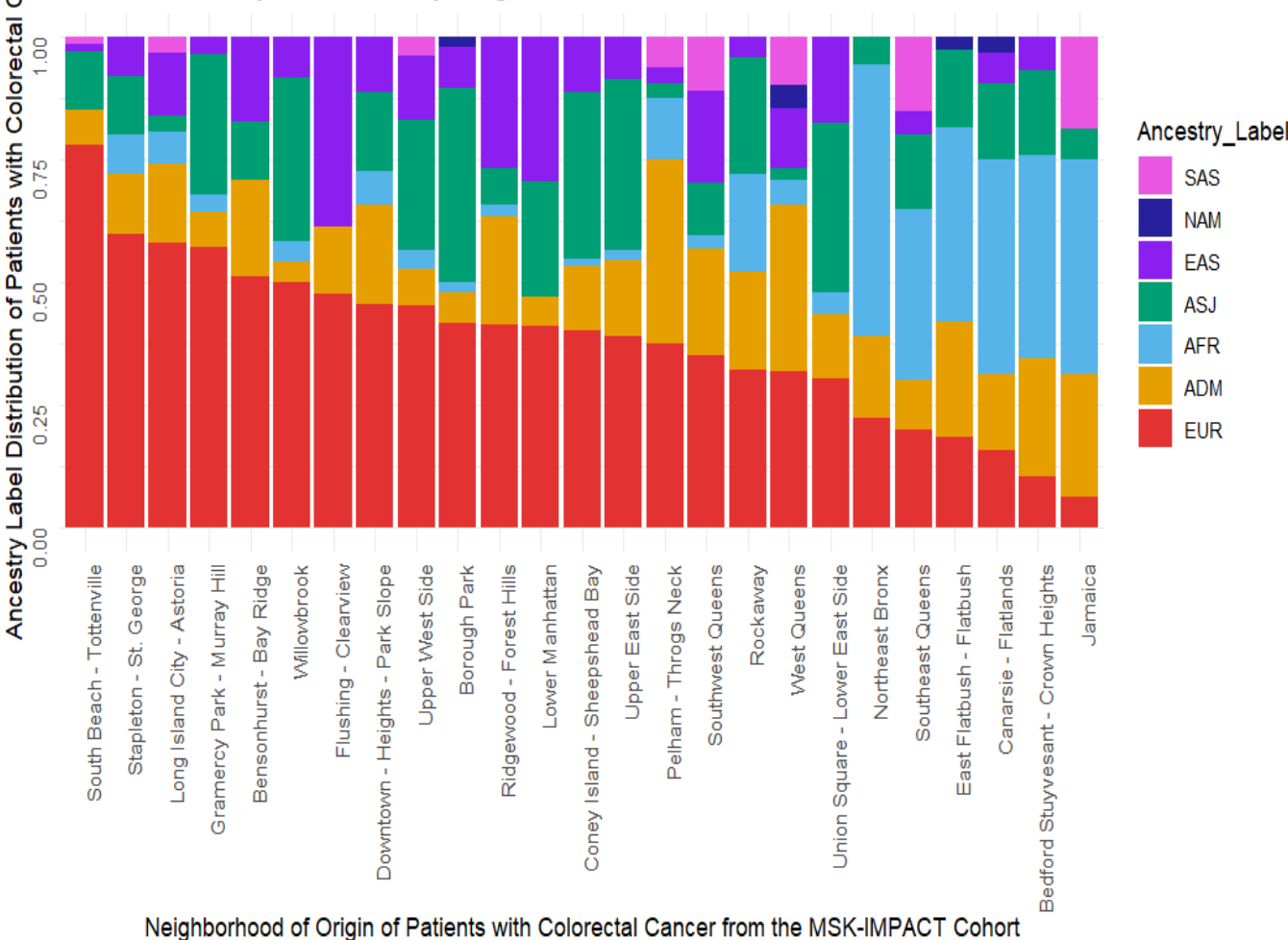
Results: The variation in Ancestry Labels were statistically significant across neighborhoods

Distribution of European Patients with Colorectal Cancer from MSK-IMPACT per NYC Neighborhood



South Beach - Tottenville

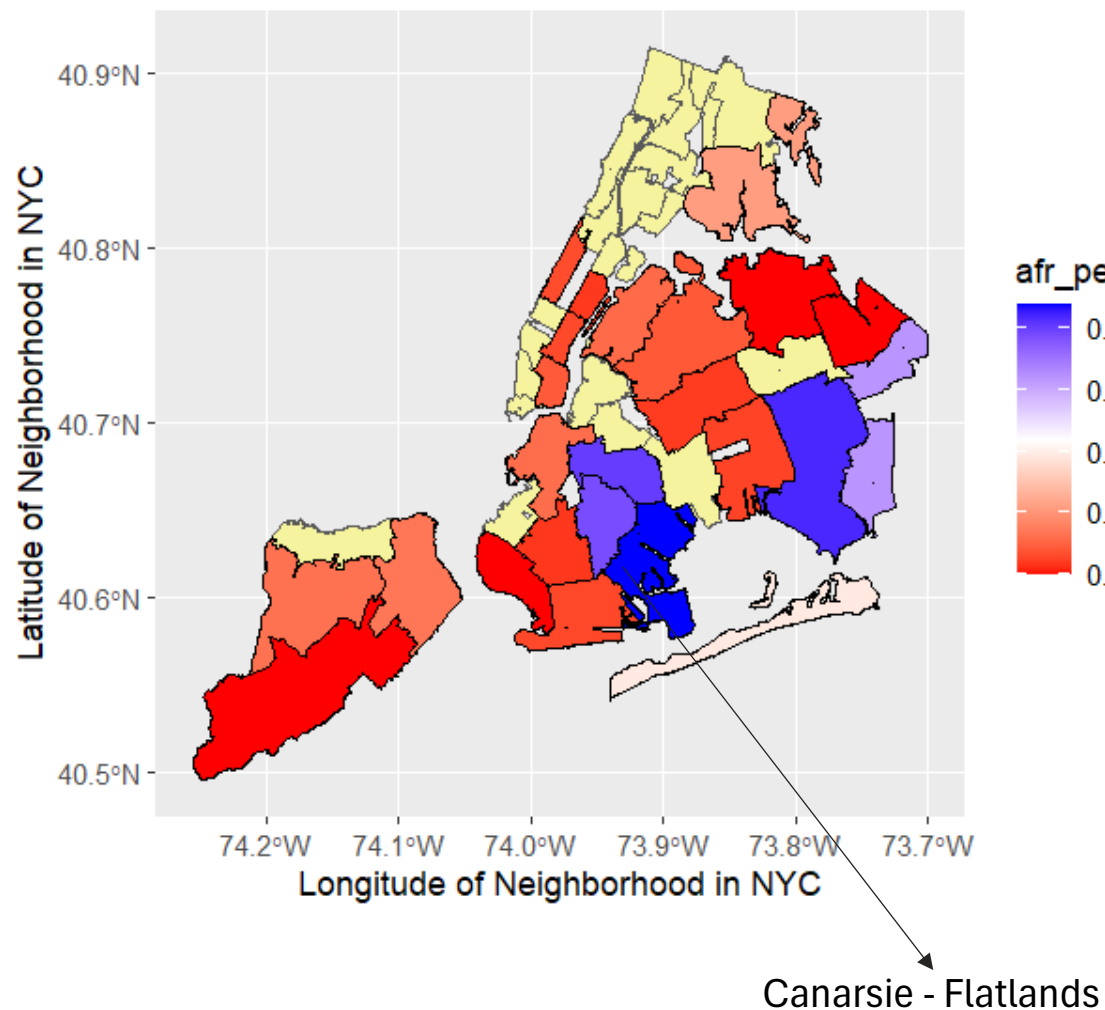
Ancestry Label Distribution of Patients with Colorectal Cancer From MSK-IMPACT by New York City Neighborhood



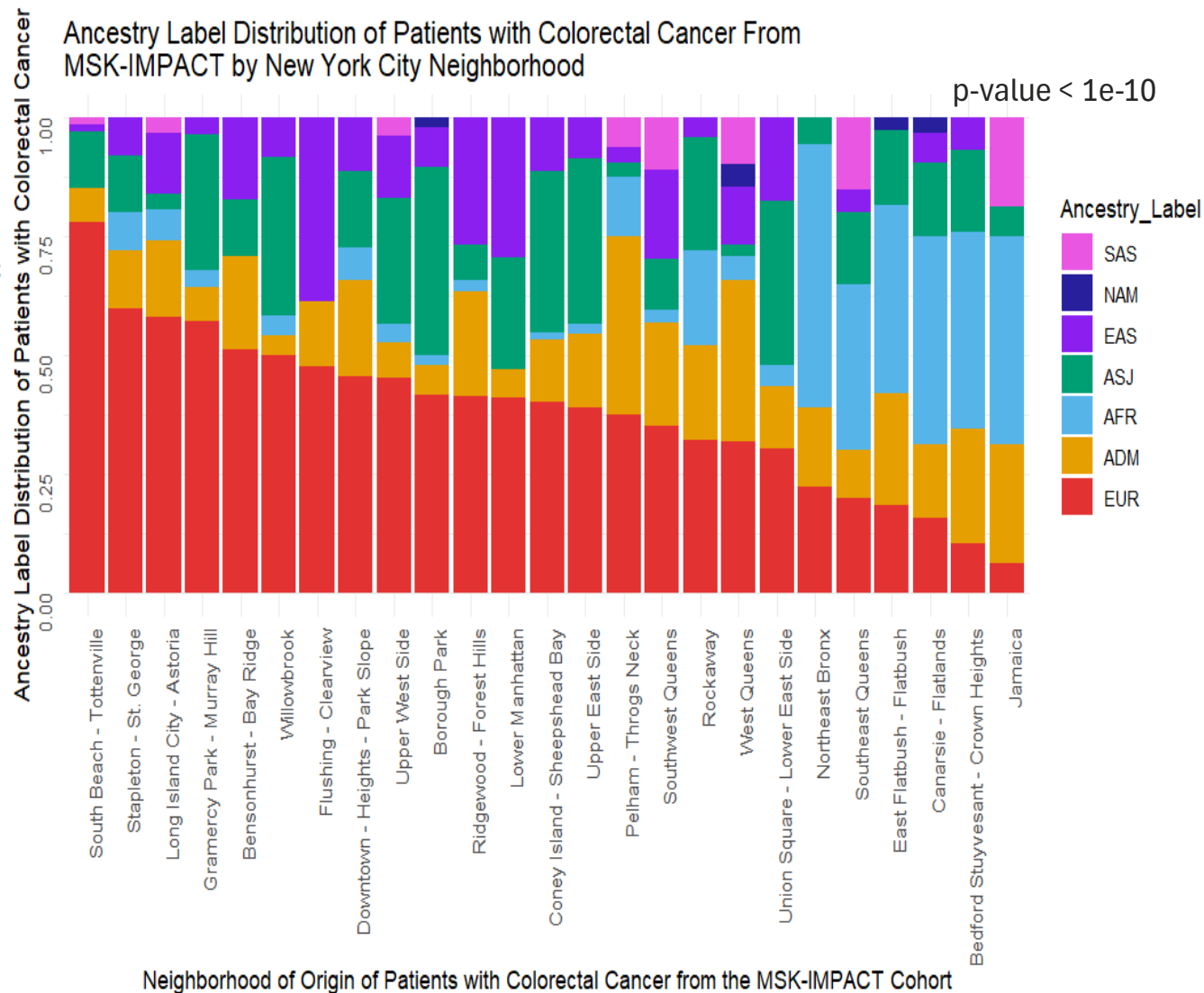
Neighborhood of Origin of Patients with Colorectal Cancer from the MSK-IMPACT Cohort

Results: The variation in Ancestry Labels were statistically significant across neighborhoods

Distribution of African Patients with Colorectal Cancer from MSK-IMPACT per NYC Neighborhood

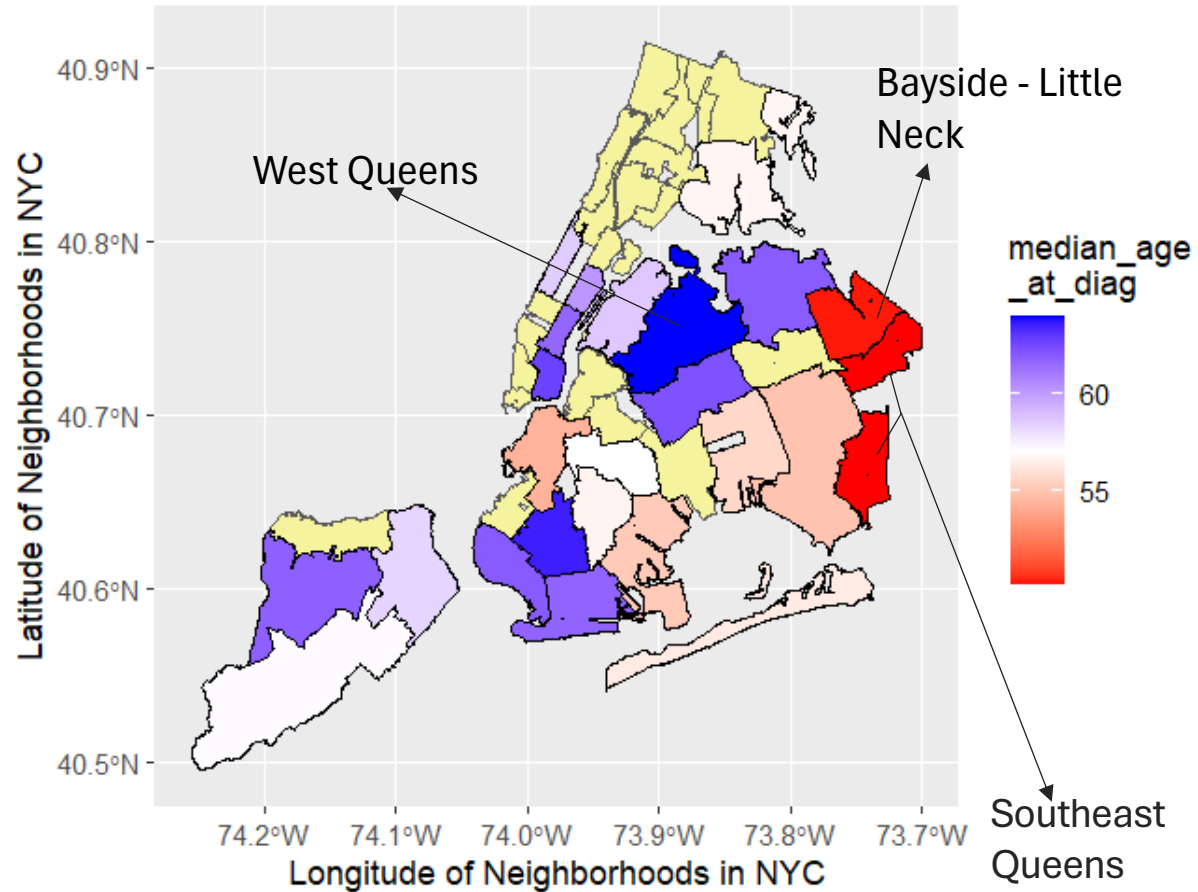


Ancestry Label Distribution of Patients with Colorectal Cancer From MSK-IMPACT by New York City Neighborhood

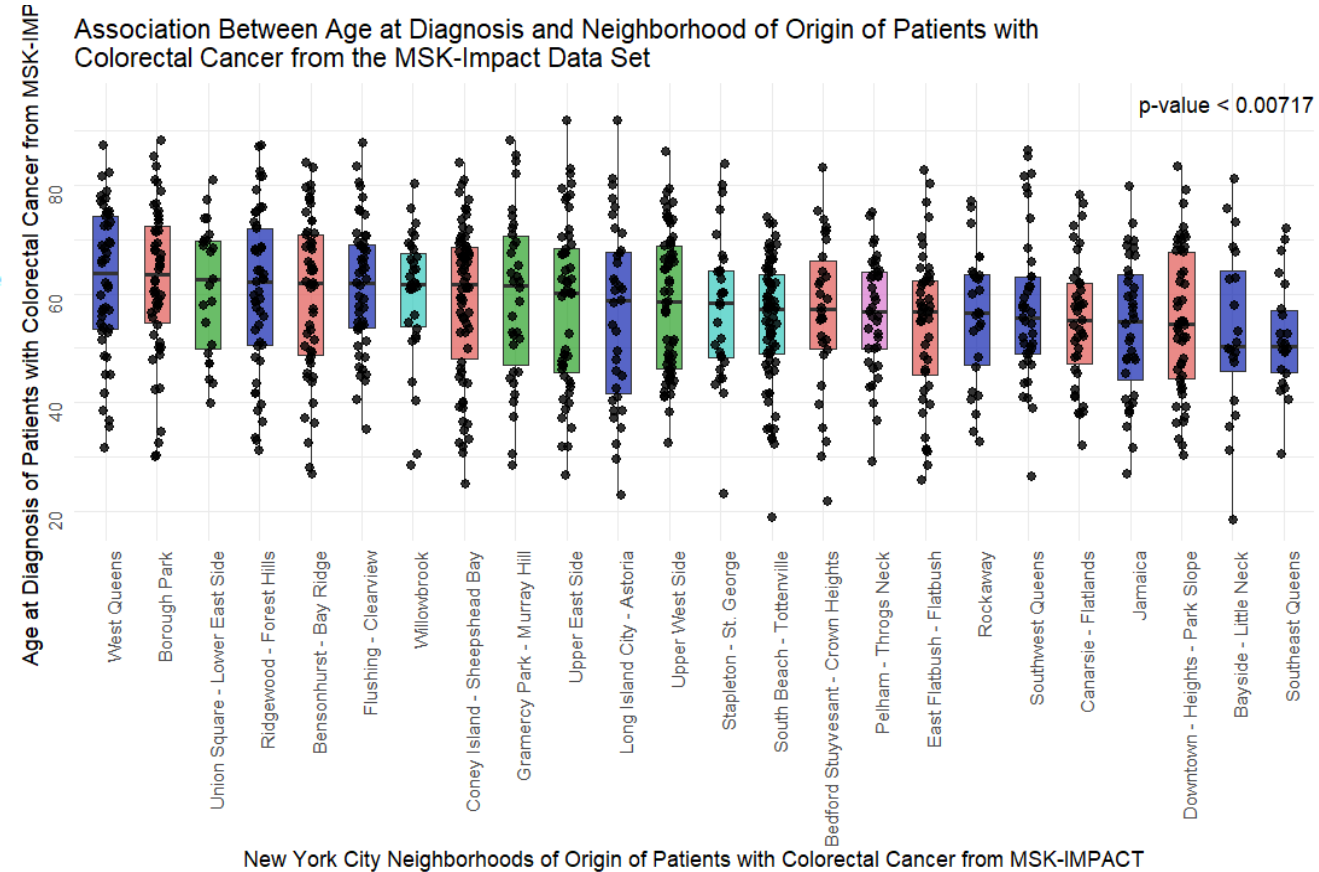


Results: The differences in Median Age at Diagnosis were statistically significant across neighborhoods

Distribution of Age at Diagnosis of Patients with Colorectal Cancer from MSK-IMPACT, per NYC Neighborhood

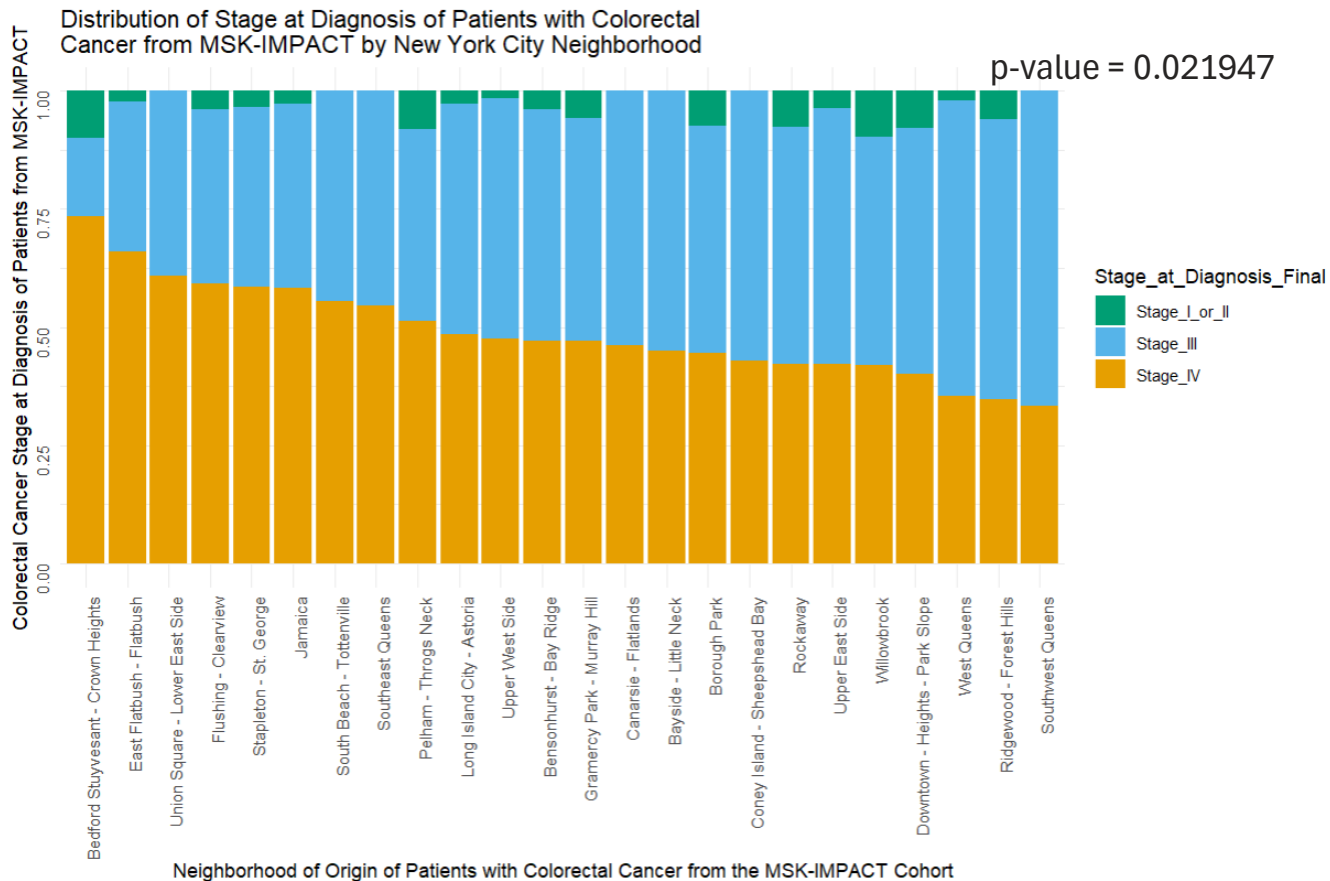
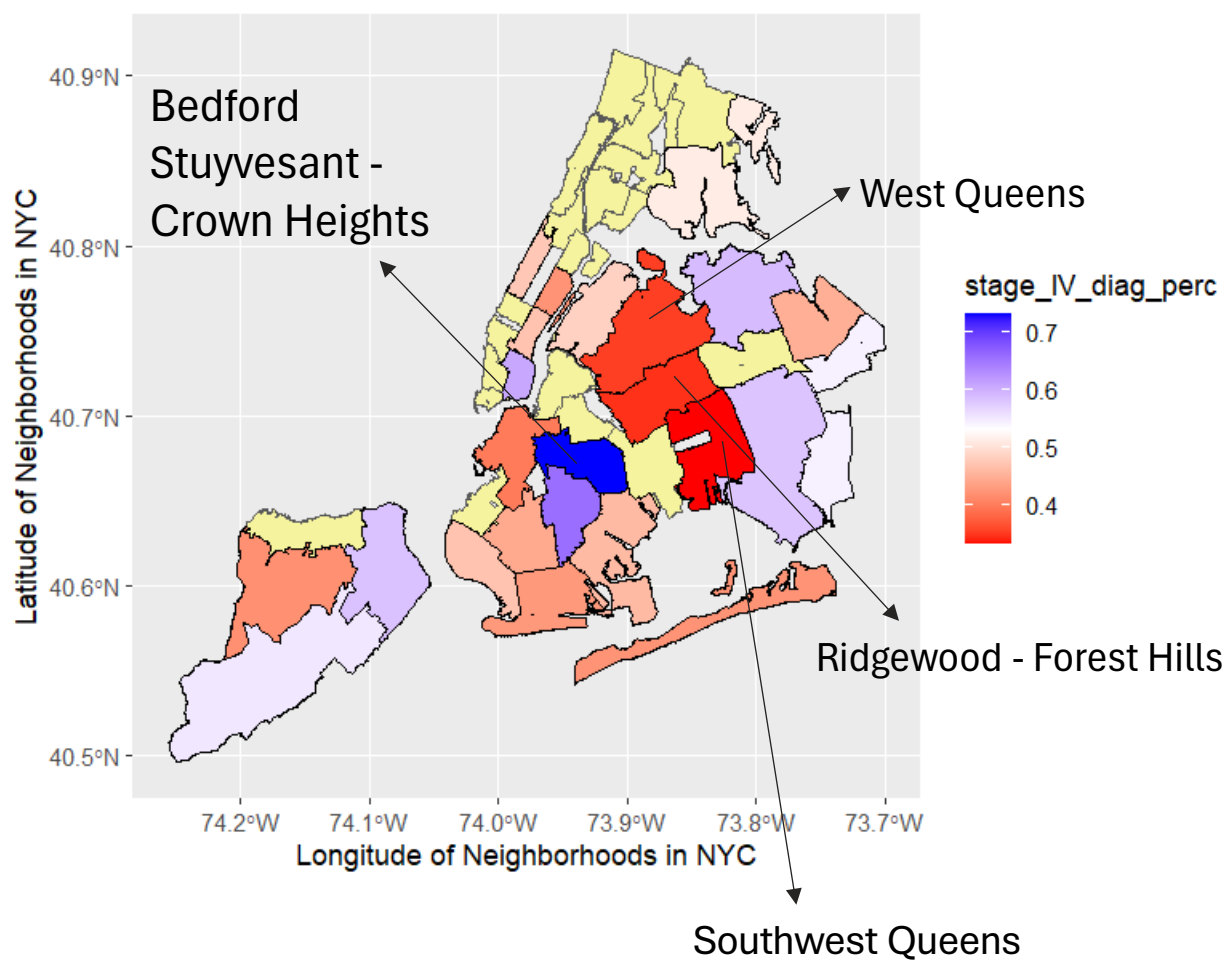


Association Between Age at Diagnosis and Neighborhood of Origin of Patients with Colorectal Cancer from the MSK-Impact Data Set



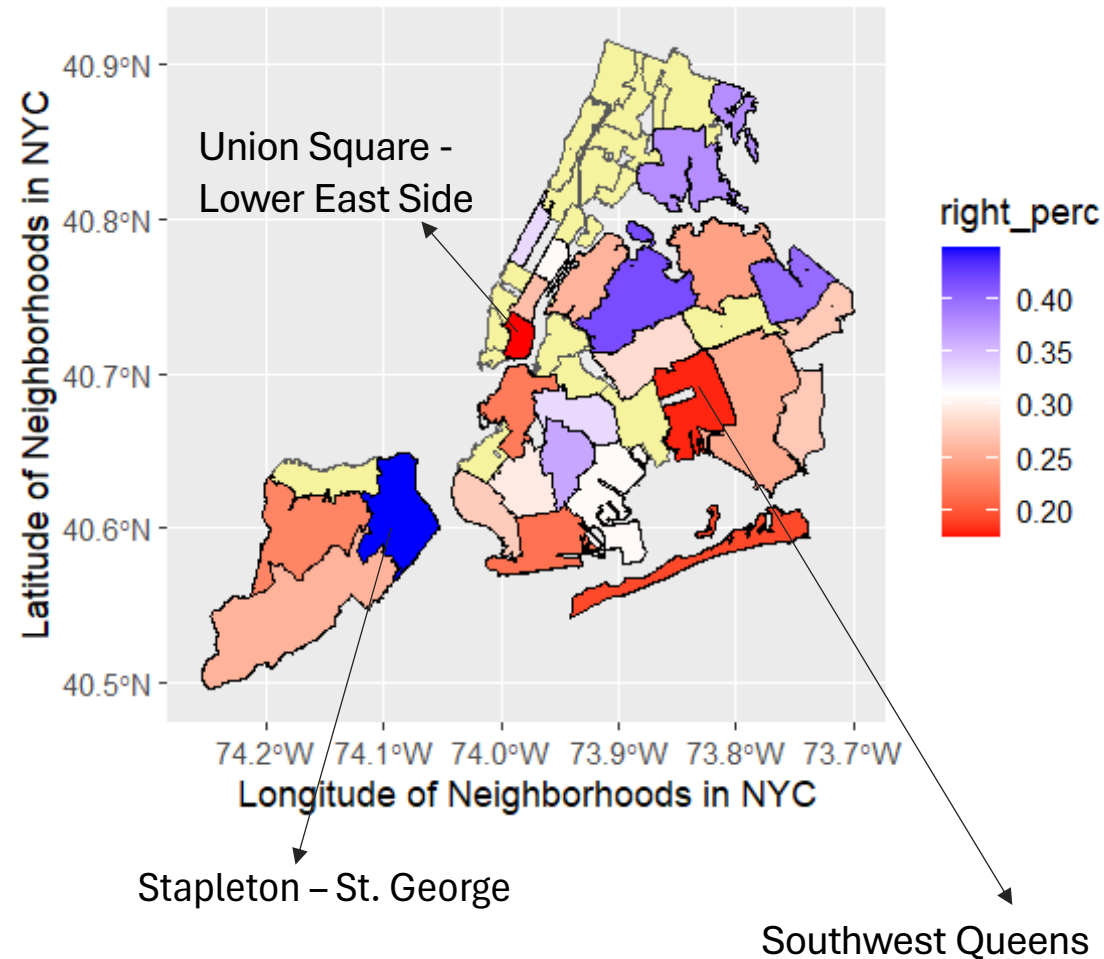
Results: The variation in Stage at Diagnosis was statistically significant across neighborhoods

Percentage of Patients from MSK-IMPACT by New York City Neighborhood with Stage IV Colorectal Cancer at Diagnosis

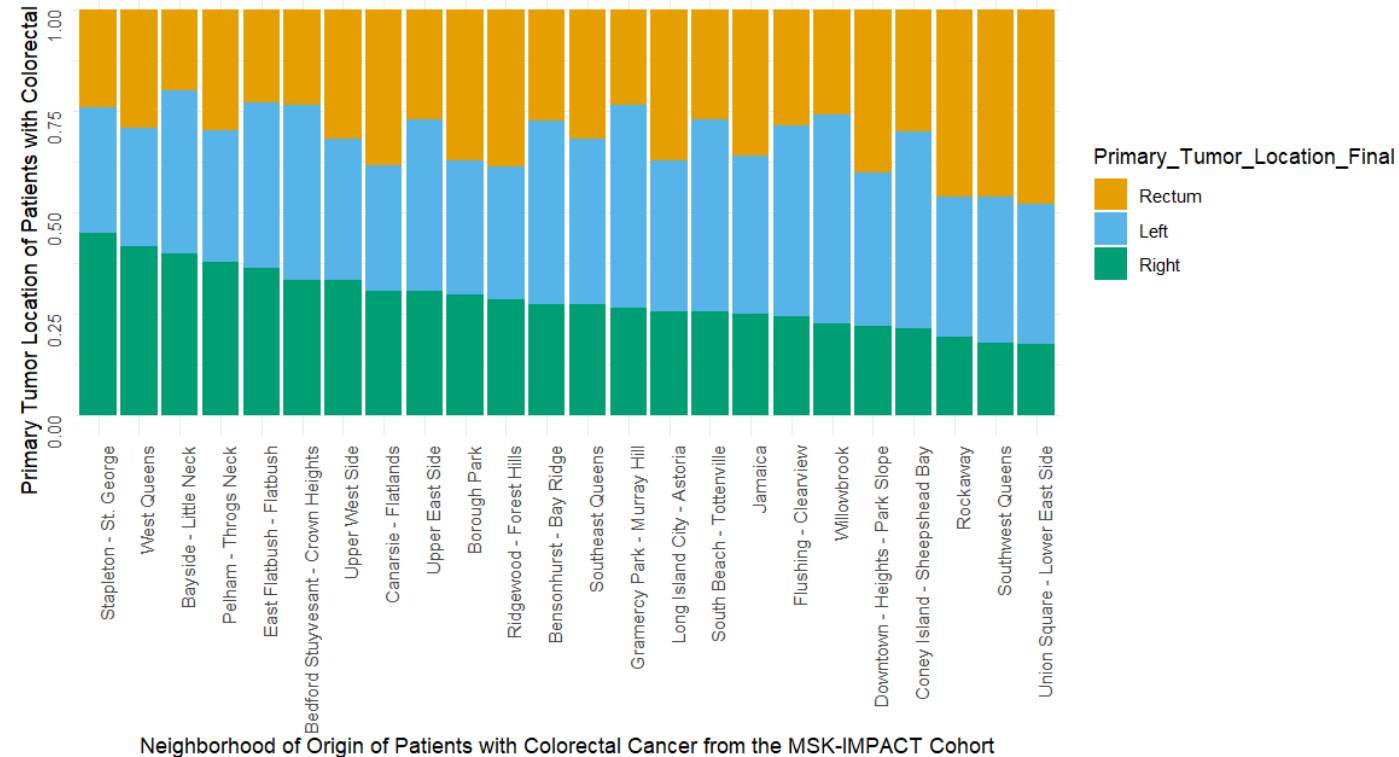


Results: There was no significant association between Primary Tumor Location and neighborhood, although some variation is still observed

Distribution of the Patients with Right-sided Colorectal Cancer from MSK-IMPACT, per NYC Neighborhood

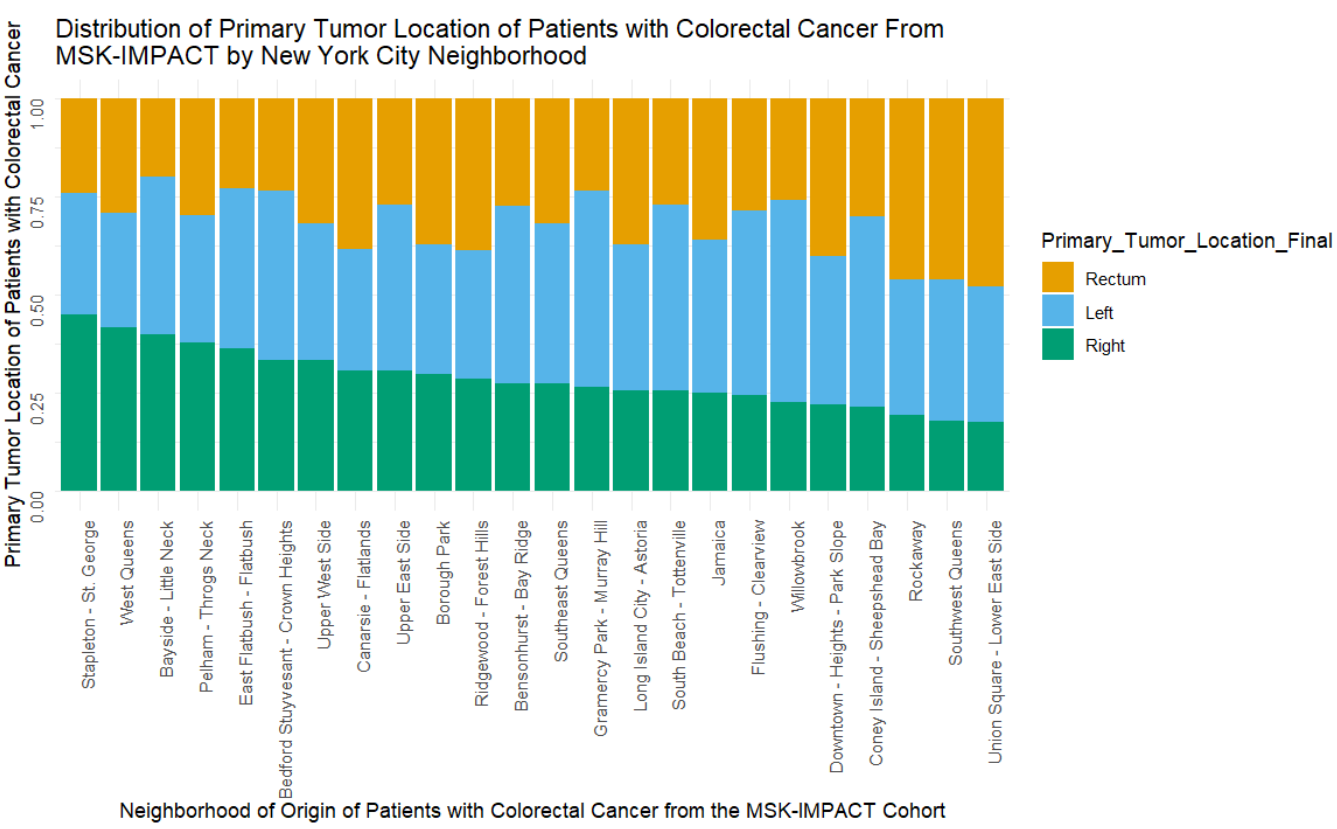
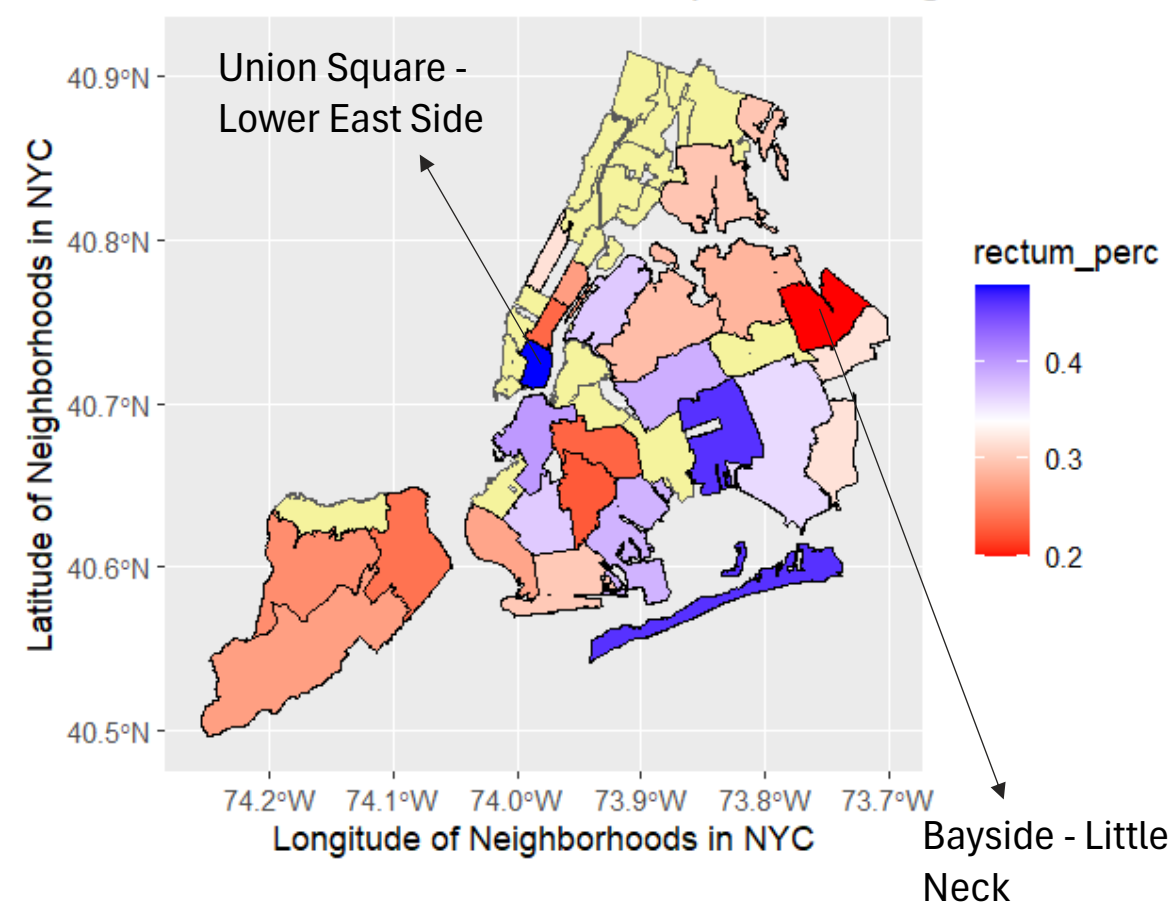


Distribution of Primary Tumor Location of Patients with Colorectal Cancer From MSK-IMPACT by New York City Neighborhood



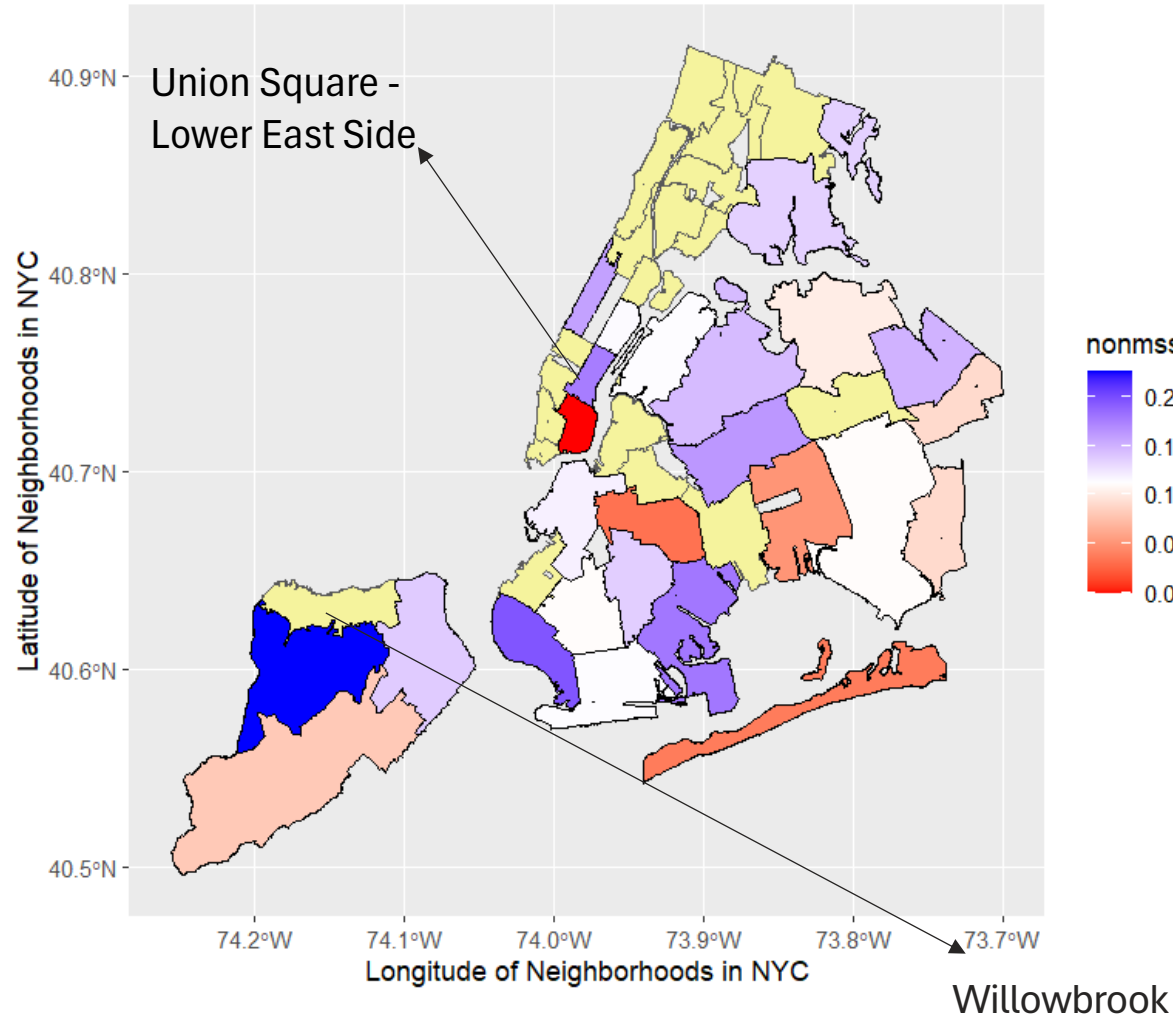
Results: There was no significant association between Primary Tumor Location and neighborhood, although some variation is still observed

Distribution of the Patients with Rectum-sided Colorectal Cancer from MSK-IMPACT, per NYC Neighborhood

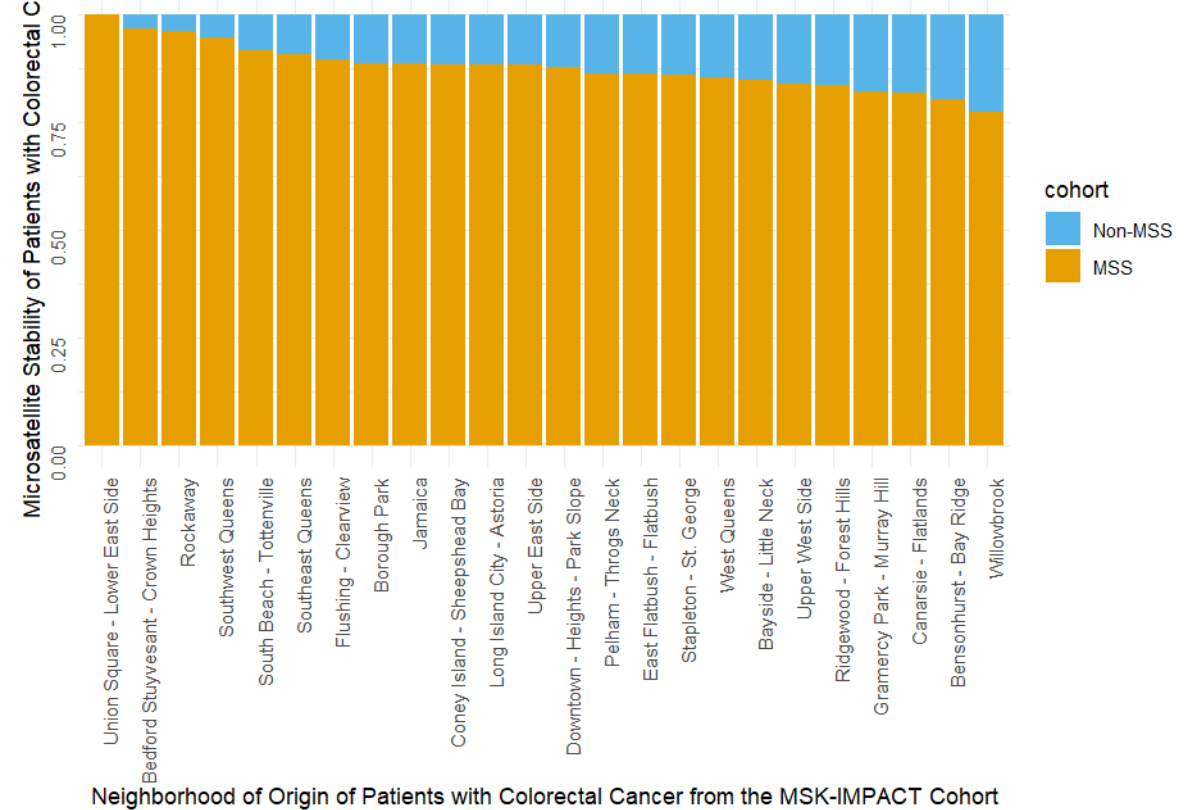


Results: There was some variation in Microsatellite Instability rates across neighborhoods, but there was no statistical significance

Distribution of Microsatellite Stability per NYC Neighborhood of Patients with Colorectal Cancer from MSK-IMPACT

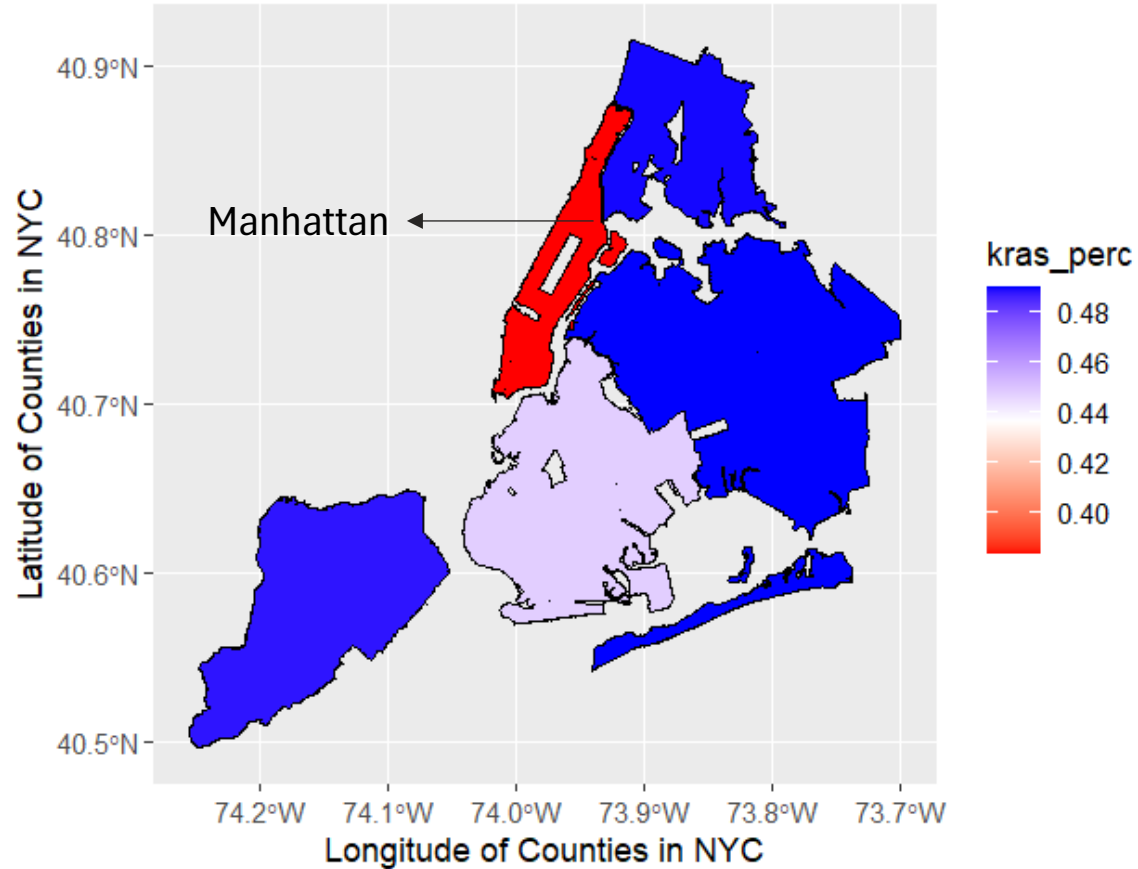


Distribution of Microsatellite Stability of Patients with Colorectal Cancer From MSK-IMPACT by New York City Neighborhood

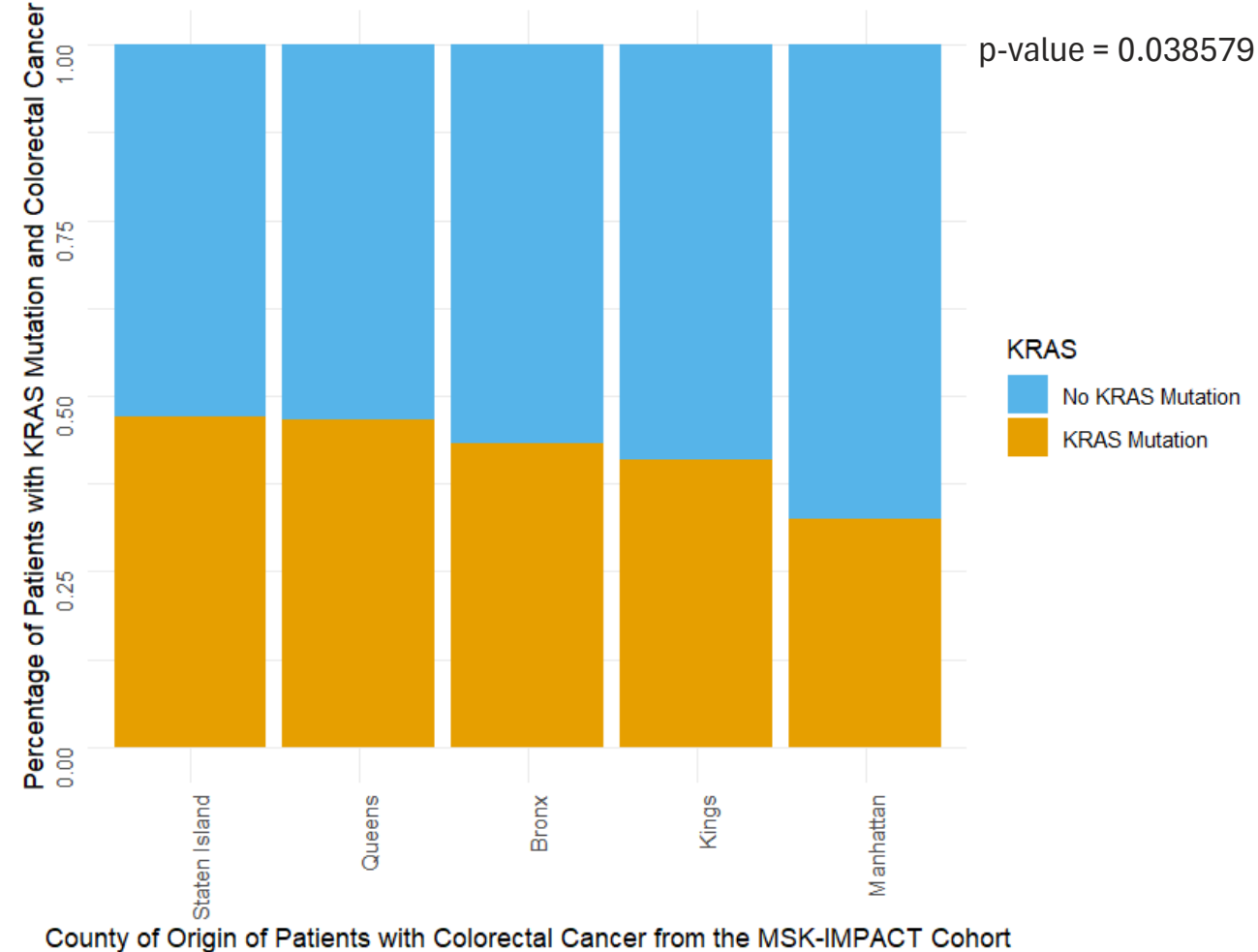


Results: The variation in KRAS mutations were statistically significant across neighborhoods

Distribution of the Patients with KRAS Mutation and Colorectal Cancer from MSK-IMPACT, per NYC County.

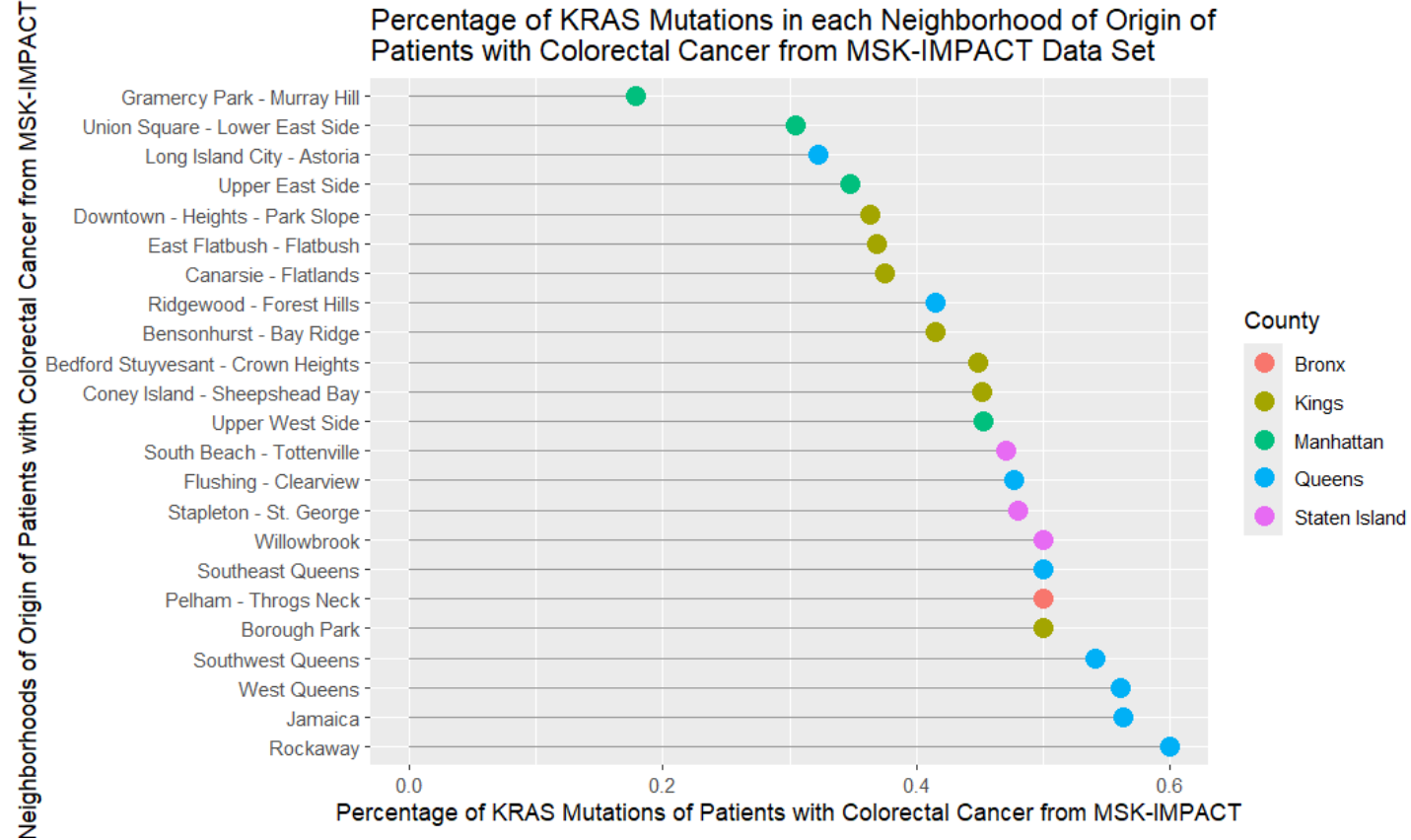
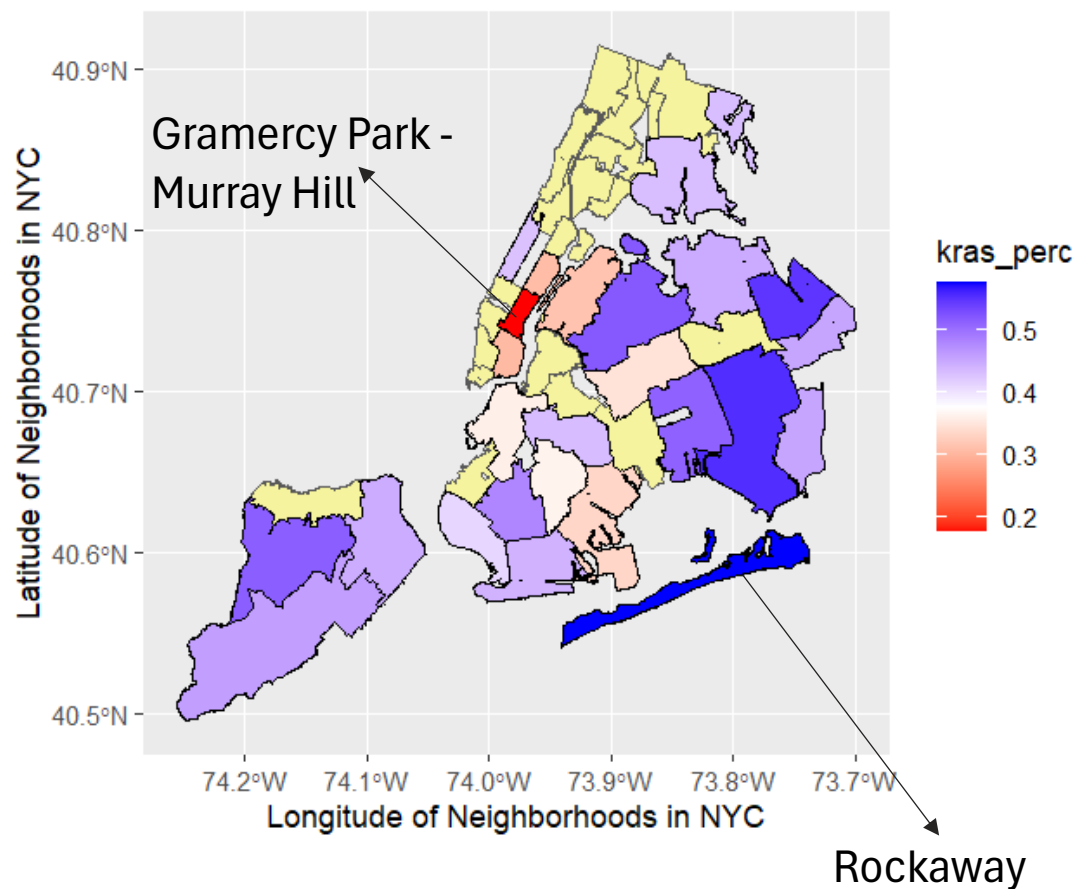


Distribution of Patients with KRAS Mutation and Colorectal Cancer From MSK-IMPACT by New York City County



Results: There are noticeably less KRAS mutations in Gramercy Park – Murray Hill than other NYC neighborhood

Distribution of the Patients with KRAS Mutation and Colorectal Cancer from MSK-IMPACT, per NYC Neighborhood.



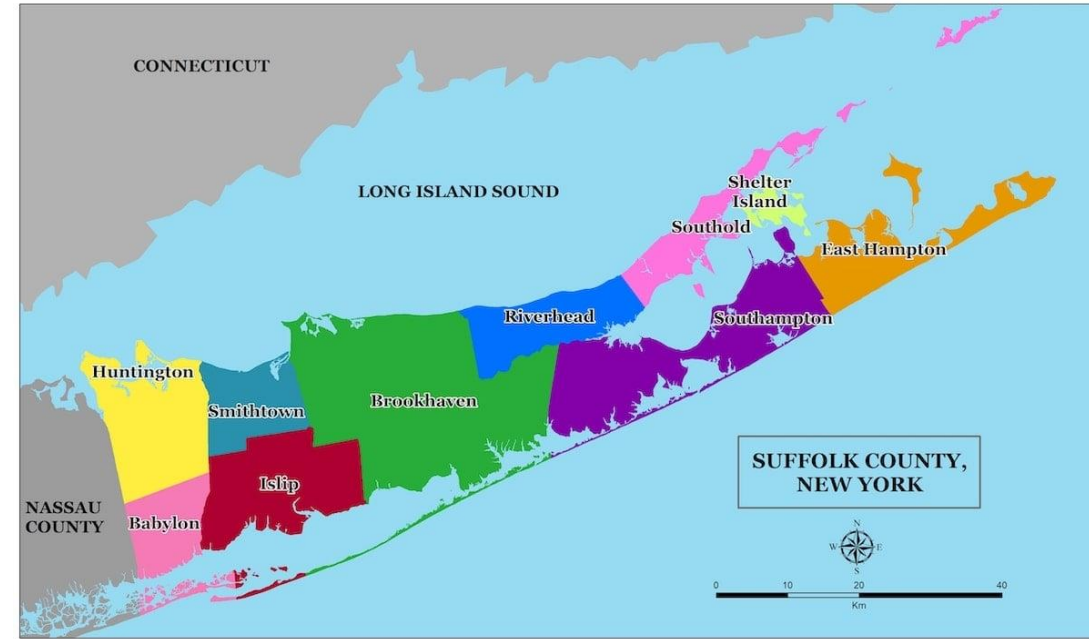
Conclusion: The clinico-genomic landscape of colorectal cancer in NYC is not geographically uniform.

- Yost Index ~ County and Neighborhood
 - Ancestry Label ~ County and Neighborhood
 - Age at Diagnosis ~ Neighborhood
 - First BMI Measurement ~ County and Neighborhood
 - Stage at Diagnosis ~ Neighborhood
 - KRAS ~ County
- Socioeconomic
- } Demographic
- } Clinical
- Genomic



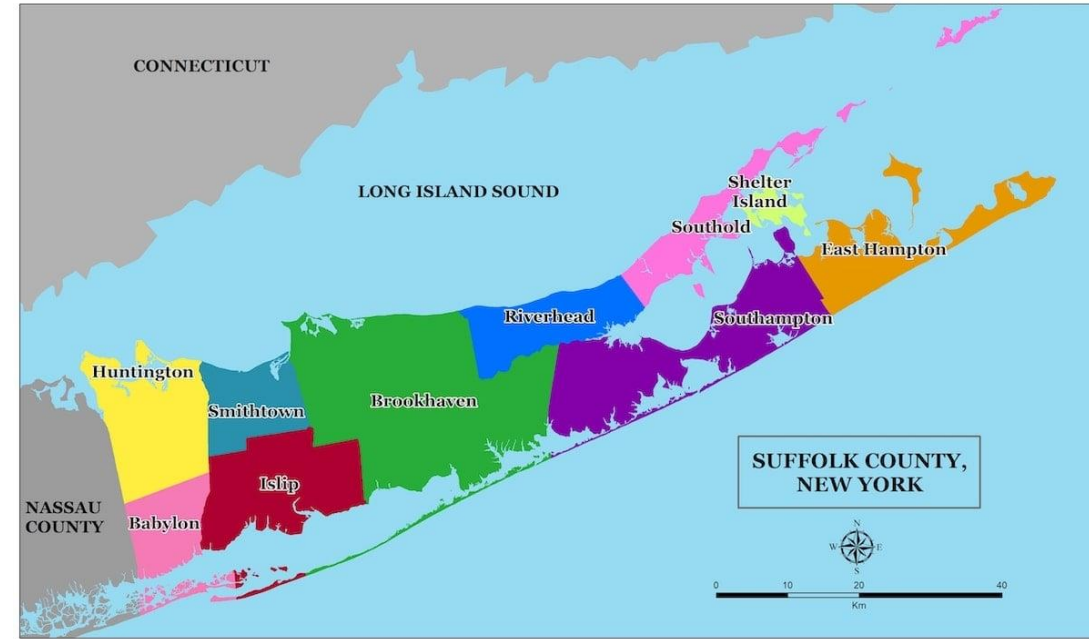
Conclusion: The next steps are a multivariate analysis, to expand to other cancer types, and to expand to other regions

- Limitations:
 - Only patients at MSK
 - Only NYC patients
 - Using last billing address on file



Conclusion: The next steps are a multivariate analysis, to expand to other cancer types, and to expand to other regions

- Limitations:
 - Only patients at MSK
 - Only NYC patients
 - Using last billing address on file
- Next steps:
 - multivariate analysis
 - expand to other regions
 - expand to other cancer types
 - spatial proximity



Conclusion: The clinico-genomic landscape of colorectal cancer in NYC is not geographically uniform.

- Yost Index ~ County and Neighborhood
 - Ancestry Label ~ County and Neighborhood
 - Age at Diagnosis ~ Neighborhood
 - First BMI Measurement ~ County and Neighborhood
 - Stage at Diagnosis ~ Neighborhood
 - KRAS ~ County
- Diagram illustrating the relationship between the variables and the categories:
- Socioeconomic (Yost Index)
 - Demographic (Ancestry Label, Age at Diagnosis)
 - Clinical (First BMI Measurement, Stage at Diagnosis)
 - Genomic (KRAS)

Thank you:

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Dr. Sharafudeen Abubakar
Dr. Michele Waters
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